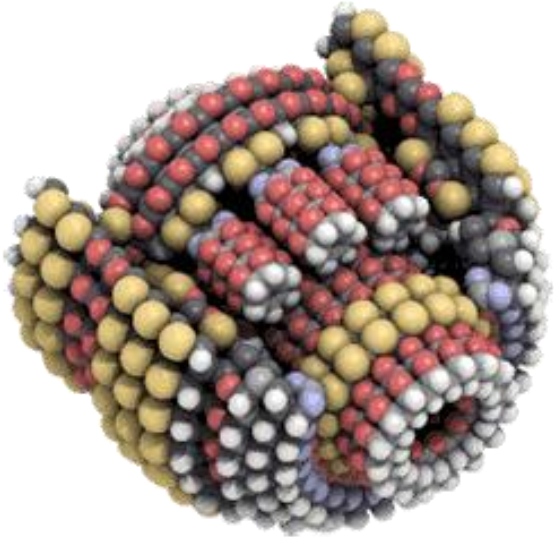
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CREATE CHANGE

MATE6301 Nanomaterials



Lecture 1: Course overview and Introduction to Nanomaterials

A/Prof. Shahriar Hossain, SoMME/EAIT
Email: md.hossain@uq.edu.au

Acknowledgement of **Country**

The University of Queensland (UQ) acknowledges the Traditional Owners and their custodianship of the lands on which we meet.

We pay our respects to their Ancestors and their descendants, who continue cultural and spiritual connections to Country.

We recognise their valuable contributions to Australian and global society.

https://www.youtube.com/watch?



Mandatory Respect at UQ module

- Commencing students enrolled at UQ in Semester 2 2025 must complete the module by **5pm on Sunday 16 August 2026.**
- Students who do not complete the module by the deadline will have their access to Blackboard removed.
- Support and adjustments to the module are available. You can find a link to the module, along with information about what's included in the module and support available at respect.uq.edu.au/respect-at-uq.



Contact details

- Lecture
  - Tuesday 14:00 – 15:50 through ECHO360 +47A-141 - Sir James Foots Building
  - Laboratory/Project/Portfolio/Presentation

Friday 8:00 – 12:00 Laboratory through recording + AIBN + CMM+ ANFF and 46-914(Seminar/discussion)

Electronic Course Profile (ECP)

https://course-profiles.uq.edu.au/course-profiles/MATE6301-60587-7660#course-overview

Teaching Staffs

Course Co-Ordinator/ Magnetic nano structure synthesis, characterisation and applications

Associate Professor
School of Mechanical and Mining Engineering
Faculty of Engineering, Architecture and Information Technology

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Nanomaterials characterisation by electron microscopy: XRD, SEM and TEM

Dr
Aditya Ashok

Email: a.ashok@uq.edu.au



Magneto-optical Nanomaterials applications: biomedical and agriculture

Postdoctoral Research Fellow
Australian Institute for Bioengineering and Nanotechnology

m.masud@uq.edu.au



Teaching Staffs

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AUSTRALIA

Nanocatalysts fabrication and characterisation for waste water treatment

Dr Valentino Kaneti

Postdoctoral Research Fellow
Australian Institute for Bioengineering and Nanotechnology

v.kaneti@uq.edu.au



Nanocellulose manufacturing, characterisation and applications

Senior Research Fellow
Australian Institute for Bioengineering and Nanotechnology

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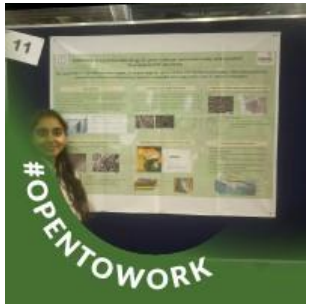


Mr. Tim Ma

Mr Mehdi Hasan



Ms Swathi



Ms Lintang



Topics/contents of the course

- Nanomaterials
  - Properties, Fabrications and Applications
  - Type of Nanomaterials
    - mesoporous gold nanostructures
    - Magnetic nanocomposites
    - Nanoporous carbon materials
    - Nanocatalysts
    - Nanocellulose
    - Metallic nano-composites
    - Semiconductor nanostructures
- Characterisation techniques
  - X-ray diffraction
  - Scanning electron microscopy
  - Transmission electron microscopy
- Applications
  - Energy, medical and environment

Highlights:

Lab pracs at AIBN, CMM, ANFF laboratories.

Weekly drop-in session, if requires. by communicating with teaching staff

Textbooks

- Textbook
  - No particular textbook recommended
- Concepts to be learnt
  - All concepts to be learnt will be outlined in the lecture notes
- References
  - Journals: *Nature*, *Science*, *Advanced Materials*, *Nano Letters*, *Materials Today* (all these journals have on-line versions at UQ)
  - Listed Reference Books
  - Recommend Internet Websites

Assessments

Assessment summary

Category	Assessment task	Weight	Due date
Portfolio	Active class/Laboratory engagement and progress portfolio	20%	13/10/2026 4:00 pm
Presentation	Critical analysis on selected published literature on nanomaterials, their characterisations and applications In-person	20%	Upload the presentation slide 21/10/2026 10:00 am Presentation 23/10/2026 8:00 am
Paper/ Report/ Annotation, Project	Development and characterisation of nanoporous gold film	20%	14/09/2026 4:00 pm
Examination	Final Exam Hurdle Identity Verified	40%	End of Semester Exam Period 7/11/2026 - 21/11/2026

Quiz to test the engagement during prac + portfolio/reflection submission (submission date has been requested to change)

Upload slide on 14/10/2026 at 10am

Presentation on 16/10/2026 at 8am to 12pm

The final exam will comprehensively test knowledge learned during the course. *A minimum achievement of 40% in the final exam is required to achieve a passing grade for the course*.

Duration: 120 minutes

Format: Short essay, Problem solving

Applications for extensions

- Individuals working in the engineering, architecture, computing, IT and design sectors are often required to submit work to strict deadlines where late submissions are not accepted.
- For this reason, and as part of preparing you to enter the professional workforce, the late penalty in this course is as follows:
- Please keep a record of your submission time.
- However, if there are medical or other acceptable reasons that will affect your ability to complete an assessment by the due date and time, then you can apply for an extension or deferred examination.

Submission guidelines

Submit via TurnItIn on Blackboard.

Deferral or extension

You may be able to apply for an extension.

The maximum extension allowed is 14 days. Extensions are given in multiples of 24 hours.

Feedback is provided to students following 14 calendar days.

A Student Access Plan (SAP) can only be used for a first extension. Extensions based on an SAP may be granted for up to seven (7) days, or the maximum number of days specified in the Electronic Course Profile (ECP), if it is less than seven (7) days. Any further extensions will require additional supporting documentation, such as a medical certificate.

Late submission

A penalty of 10% of the maximum possible mark will be deducted per 24 hours from time submission is due for up to 7 days. After 7 days, you will receive a mark of 0.

https://course-profiles.uq.edu.au/course-profiles/MATE6301-60667-7460#assessment

For example:

4 marks (2 days late x 10% x 20 marks).

2 hours over 24 hrs (26 hours) delay counts as 2 days

Academic integrity and plagiarism

- Academic Integrity
  - It is the University's task to encourage ethical scholarship and to inform students and staff about the institutional standards of academic behaviour expected of them in learning, teaching and research. Students have a responsibility to maintain the highest standards of academic integrity in their work. Students must not cheat in examinations or other forms of assessment and must ensure they do not plagiarise.
- Plagiarism
  - Plagiarism is the act of misrepresenting as one's own original work the ideas, interpretations, words or creative works of another. These include published and unpublished documents, designs, music, sounds, images, photographs, computer codes and ideas gained through working in a group. These ideas, interpretations, words or works may be found in print and/or electronic media.
  - Students are encouraged to read the UQ Student Integrity and Misconduct policy (http://ppl.app.uq.edu.au/content/3.60.04-student-integrity-and-misconduct) which makes a comprehensive statement about the University's approach to plagiarism, including the approved use of plagiarism detection software, the consequences of plagiarism and the principles associated with preventing plagiarism.
  - https://web.library.uq.edu.au/library-services/it/learnuq-blackboard-help/academic-integrity-modules

Assessments (Grades v Percentages)

Grade 7 = 85-100

Grade 6 = 75-84

Grade 5 = 65-74

Grade 4 = 50-64

Consider 40% hurdle

Grade 3 = 45-49

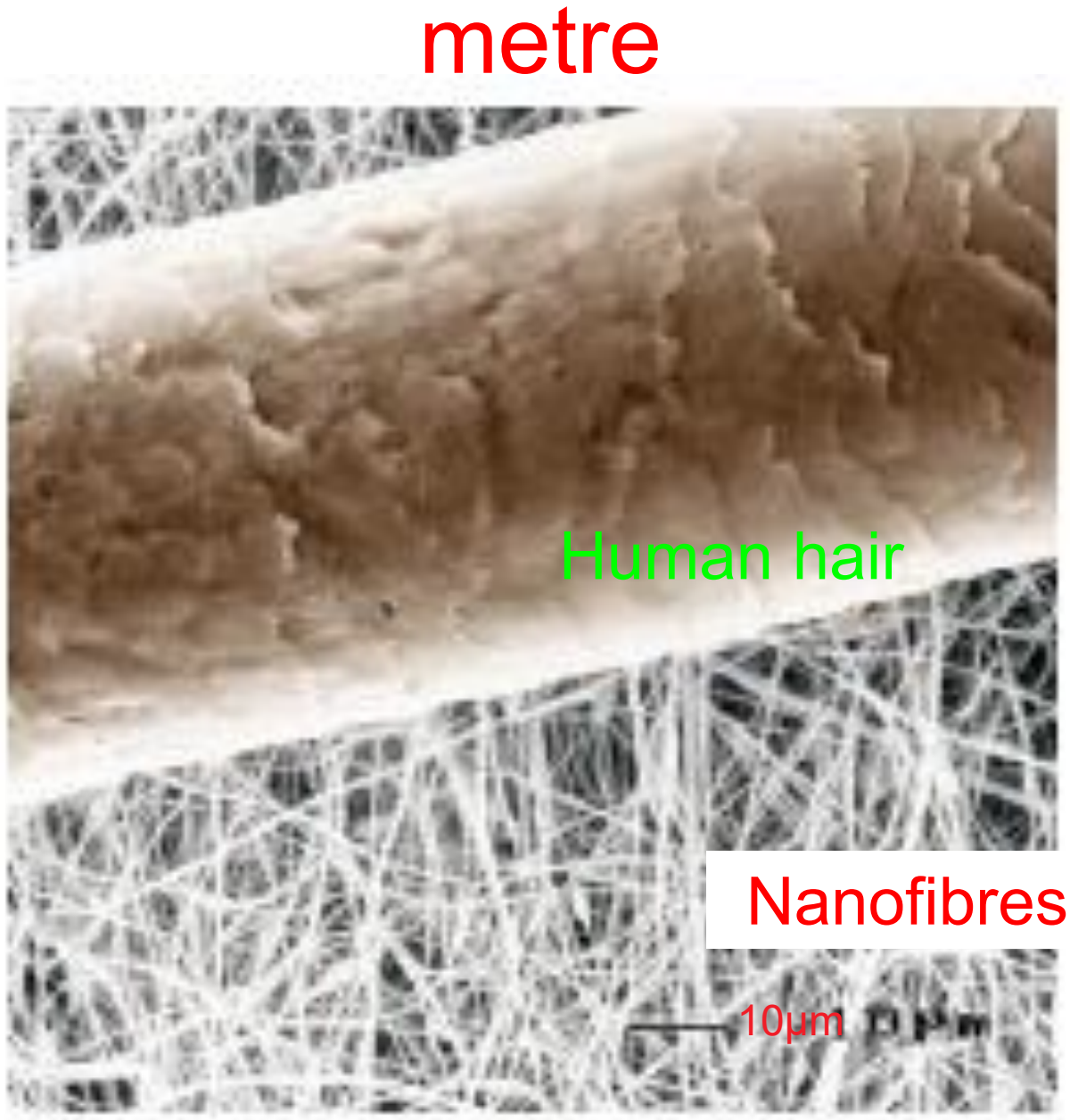
Grade 2 = 20-44

Grade 1 = 1-19

Failed

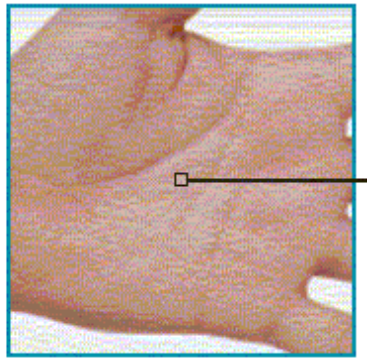
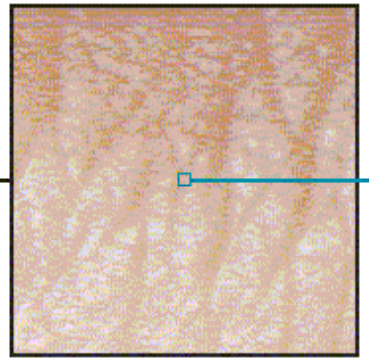
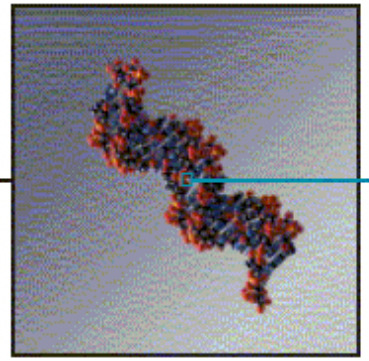
The prefixes for a given unit

Exa-	10^{18}	quintillion
Peta-	10^{15}	quadrillion
Tera-	10^{12}	trillion
Giga-	10^{9}	billion
Mega-	10^{6}	million
Kilo-	10^{3}	thousand
	1	
Milli-	10^{-3}	thousandth
Micro-	10^{-6}	millionth
Nano-	10^{-9}	billionth
Pico-	10^{-12}	trillionth
Femto-	10^{-15}	quadrillionth
Atto-	10^{-18}	quintillionth

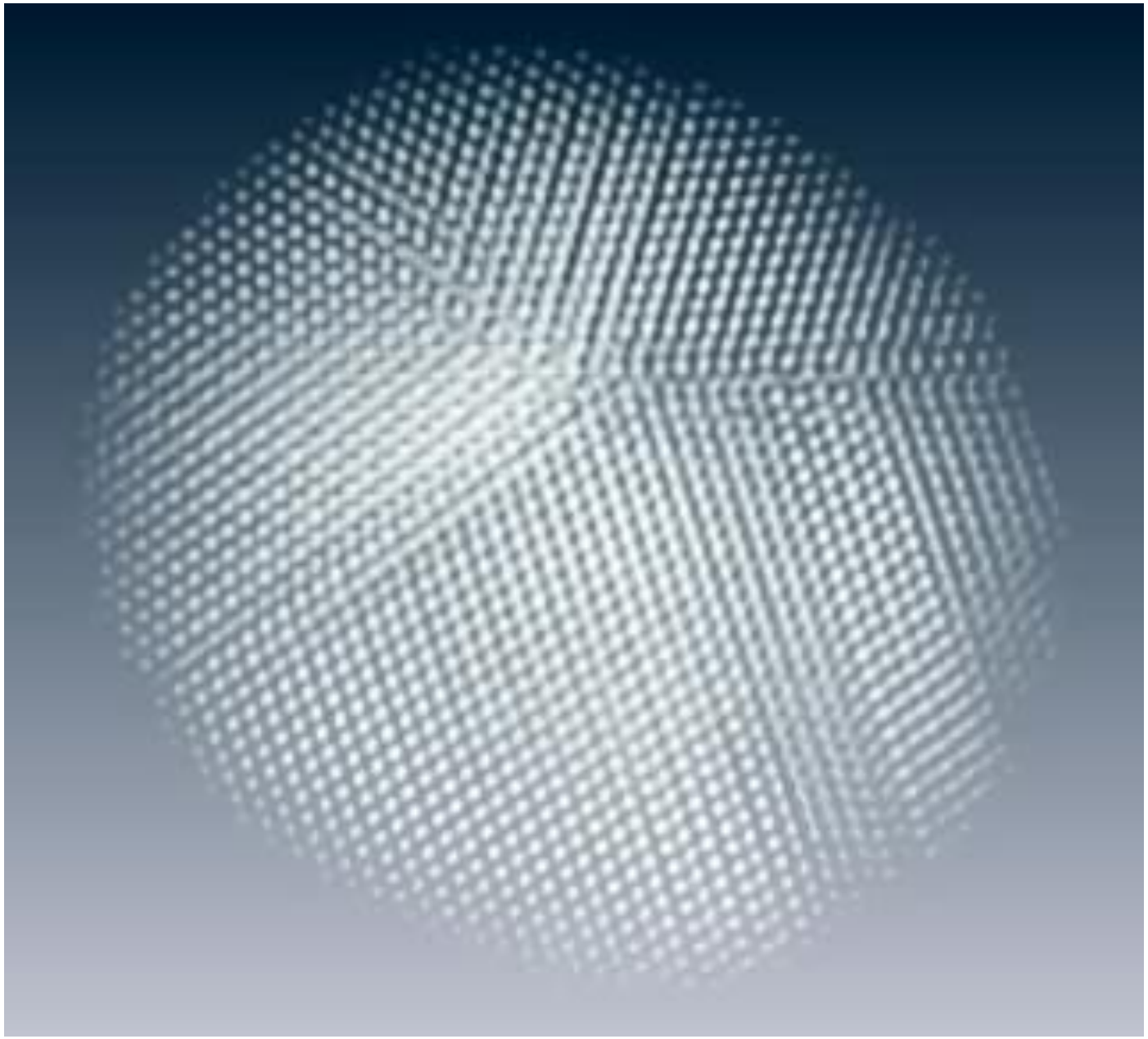


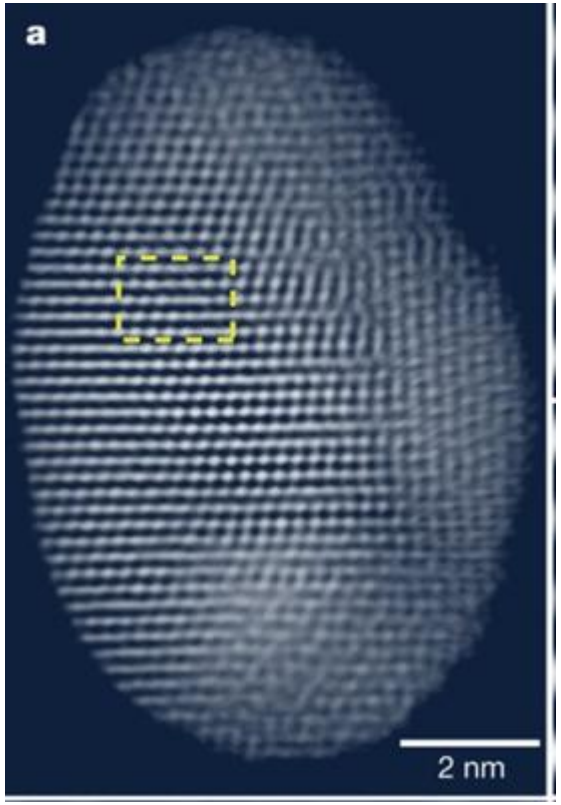
Size matters

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Hand	Skin	White Blood Cell	DNA	Atoms
				
10 cm	1 mm	10 microns	100 nanometers	1 nanometer

Size matters





A 7.9-Å-thick internal slice of the Pt nanoparticle. The lattice structure on the left and at the bottom parts of the slice is not well defined, mainly because this decahedral multiply twinned nanoparticle consists of five grains with different orientations.

Chen, CC., Zhu, C., White, E. *et al.* Three-dimensional imaging of dislocations in a nanoparticle at atomic resolution. *Nature* **496,** 74–77 (2013). https://doi.org/10.1038/nature12009

Predictions

Nano-Pioneers



Richard Feynman *(1918-1988) who won the 1965 Nobel Prize in Physics for his fundamental work in quantum electro-dynamics, in particular in 1948, he introduced so called Feynman diagrams, a graphical representation of interactions between different particles.*

The science of building small was first introduced in **1959** by Richard P. Feynman in a lecture titled ***There's Plenty of Room at the Bottom:***

"… ultimately---in the great future---we can arrange the atoms the way we want; the very atoms, all the way down! What would happen if we could arrange the atoms one by one the way we want them …"



Richard Smalley *(1943 - 2005) who won the 1996 Nobel Prize in Chemistry for his discovery of fullerenes in 1985.*

In a lecture *Nanotechnology and the Next 50 Years*, Smalley predicted: With the right advances in technology, a nanometre-sized solar cell could be built. Such devices could potentially provide for the world's energy needs in the year 2050.

Policy Maker



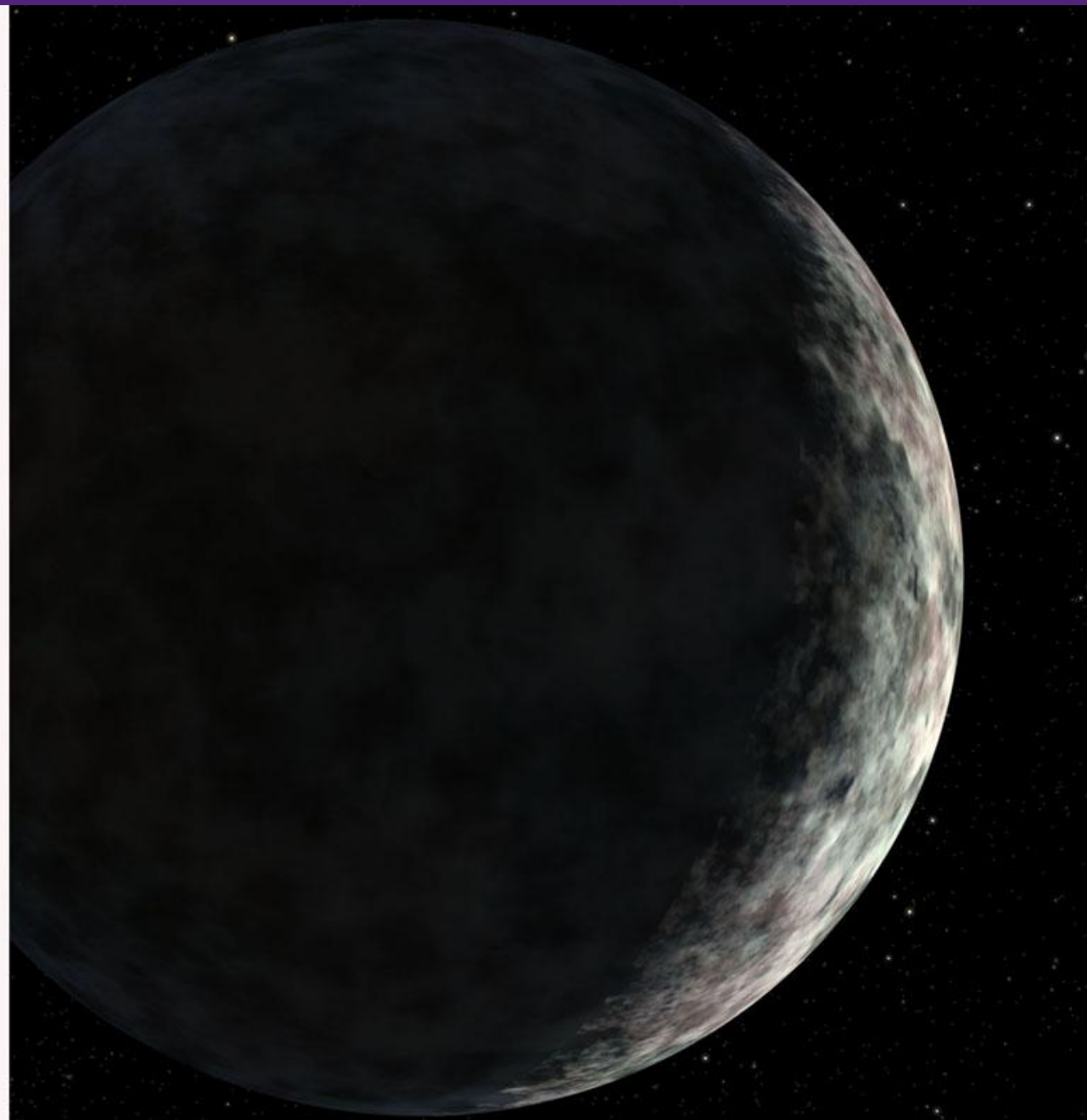
Neal Lane *(1938 -), a physicist, is the Science Advisor to former President Clinton and a former director of the National Science Foundation.*

"If I were asked for an area of science and engineering that will most likely produce the breakthroughs of tomorrow (21st Century), I would point to nanoscale science and engineering."

Global Challenges: New solutions for new challenges

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- Energy
- Water
- Food
- Environment
- Disease
- Terrorism/war
- Education
- Crime
- Poverty
- Population



Nanoscience and nanotechnology

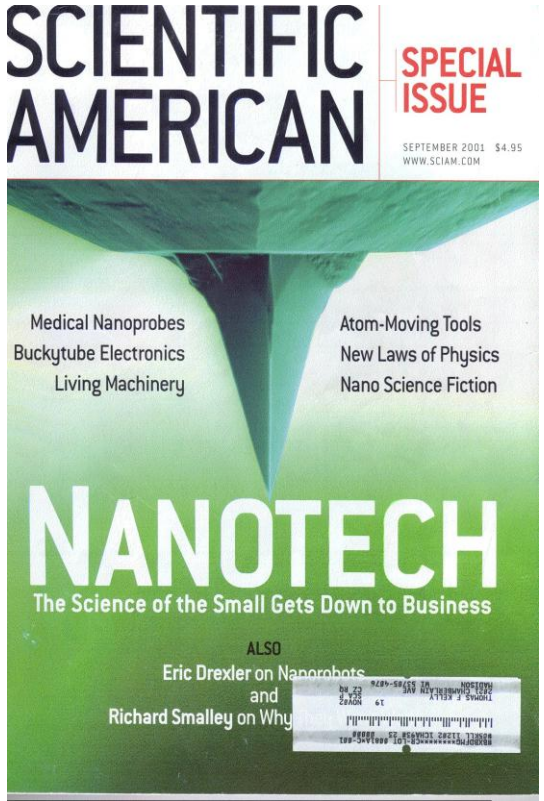
- **Nanoscience (know why)**: Study of the unique properties that emerge when system sizes approach nanometre-scale dimensions.
- **Nanotechnology(know how)**: The creation of functional/smart materials, devices, and systems through control of matter at the scale of 1 to 100 nanometres, and the exploitation of phenomena at this scale.
- **Nanoscience and nanotechnology encompasses:**
  - Sciences (Chemistry, Physics, Biology, Materials, Medicine)
  - Engineering (Mechanical, Chemical, Materials, Environmental, Metallurgical, Electrical, Biological, …)

Why nanoscience

- Small structures exhibit novel and unique properties, due to their significant fractions of surfaces/interfaces.
- Nanoscience is to understand their fundamentals
  - How they form?
  - Why do they have certain properties?
- Explore material properties and device functions beyond those that we presently know or even consider possible.
- Nanoscience is therefore an exciting new field that brings together the traditional disciplines of physics, chemistry, biology, materials science and engineering.

Why nanomaterials

- Ability to manipulate, control and build materials at the atomic and molecular level
- Provide novel affinity, capacity, and selectivity because of their unique physical, chemical and biological properties.







Nanomaterials

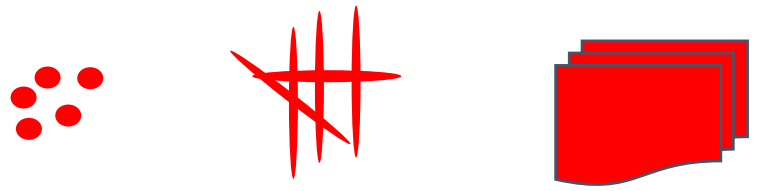
Nanomaterials stands for the materials that have a size between 1 – 100 nm.

Geometrically, nanomaterials may be considered as low (< 3) dimensional materials.

Because of their tiny in size, nanomaterials has their own characteristics.

- *Small*
- *Lightweight*
- *Novel properties*
- *Multifunctional*
- *Hierarchical*
- *Smart*

Classification

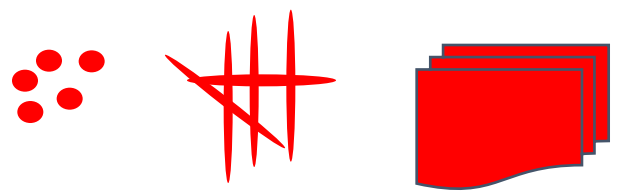


0D 1D 2D

Nanomaterials and nanostructured materials

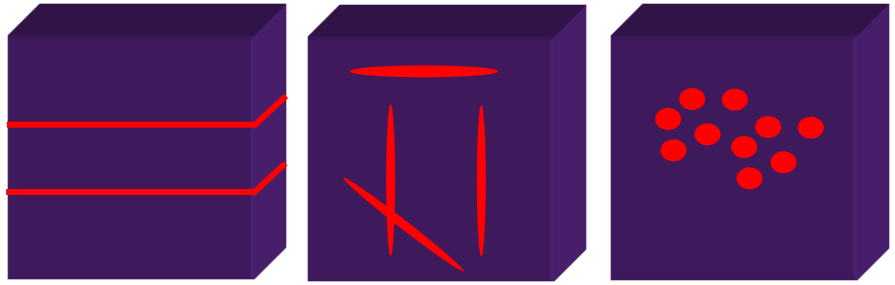
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Nanomaterials



0D 1D 2D

Nanostructured Materials



2D 1D 0D

- Nanostructured materials are bulk materials with modulated structures in 0 to 3 dimensions...
- Size constraint (*confinement*) < 100nm at least in one dimension...
- Significant volume fraction (>1%) of interfaces.

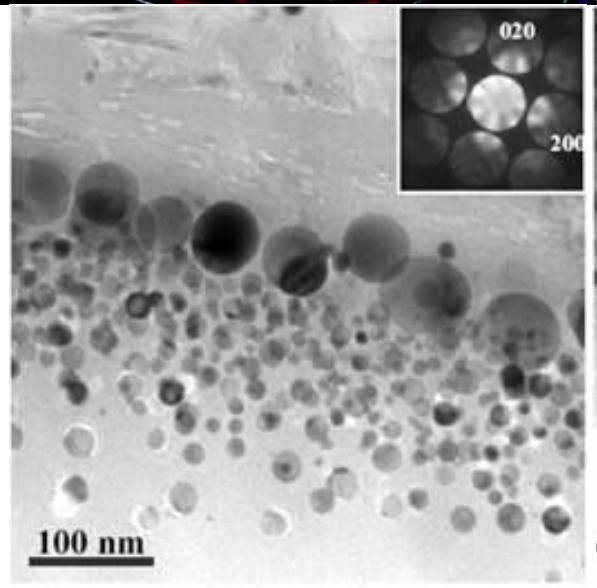
Nano is not new!

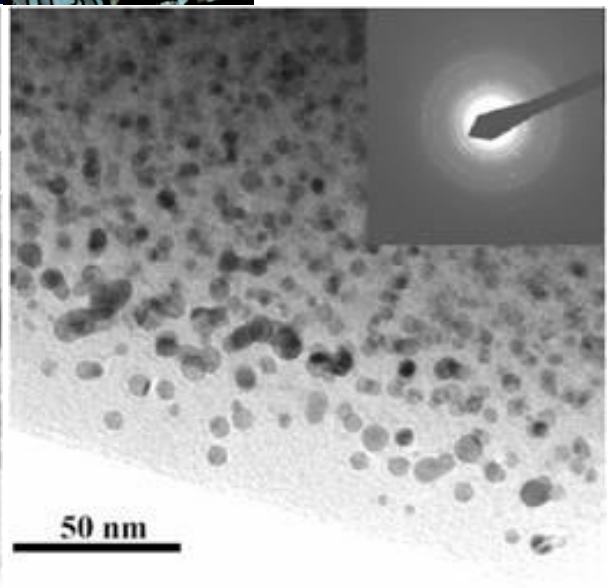


The decorative glaze, found on some medieval pottery, contains metallic nanoparticles, which gives rise to its special optical properties.



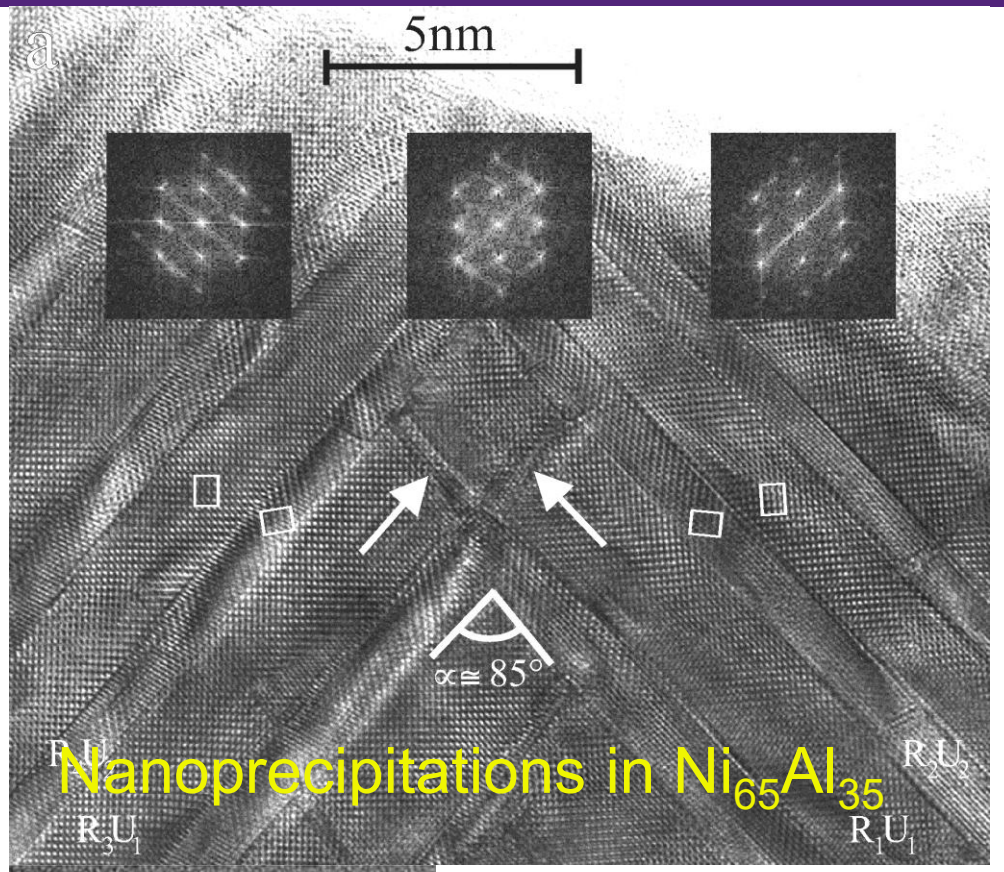
Traditionally, stained glass was made from gold nanoparticles.



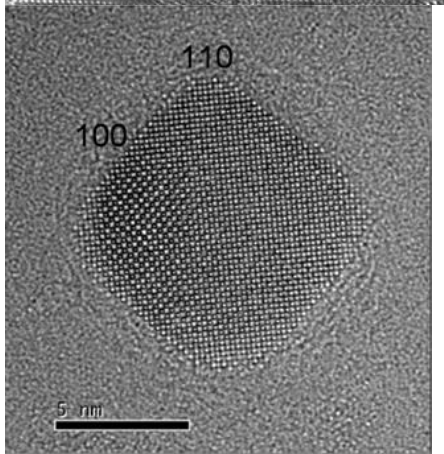


Nanomaterials are not new

- Catalysts, carbon black, sols, fumed silica and pigment industries have already commercialized several simple oxide nanopowders, nanoscale dispersion products, and nanoparticle manufacturing processes.
- Using nanosized precipitates in alloys to increase mechanical strength was developed over half a century ago.



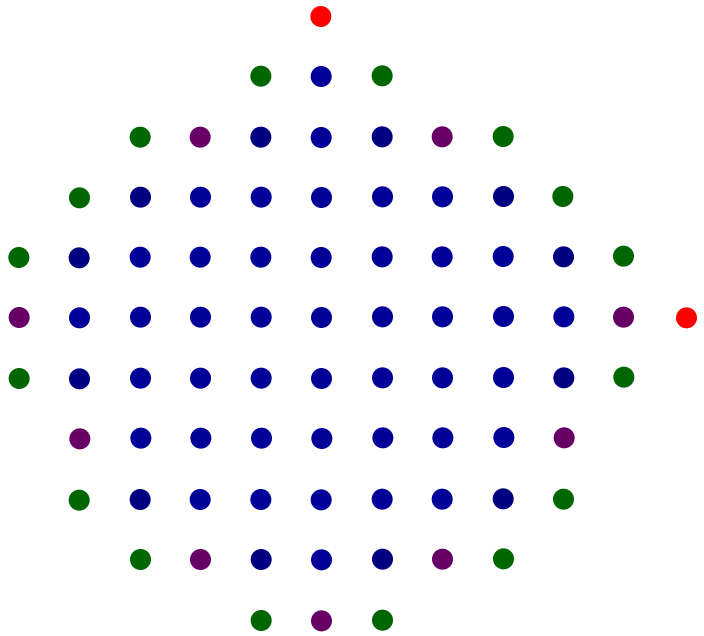
Nanoprecipitations in $Ni_{65}Al_{35}$



CeO_2 catalysts nanoparticle

Key characteristic of nanomaterials

Significant surface atoms

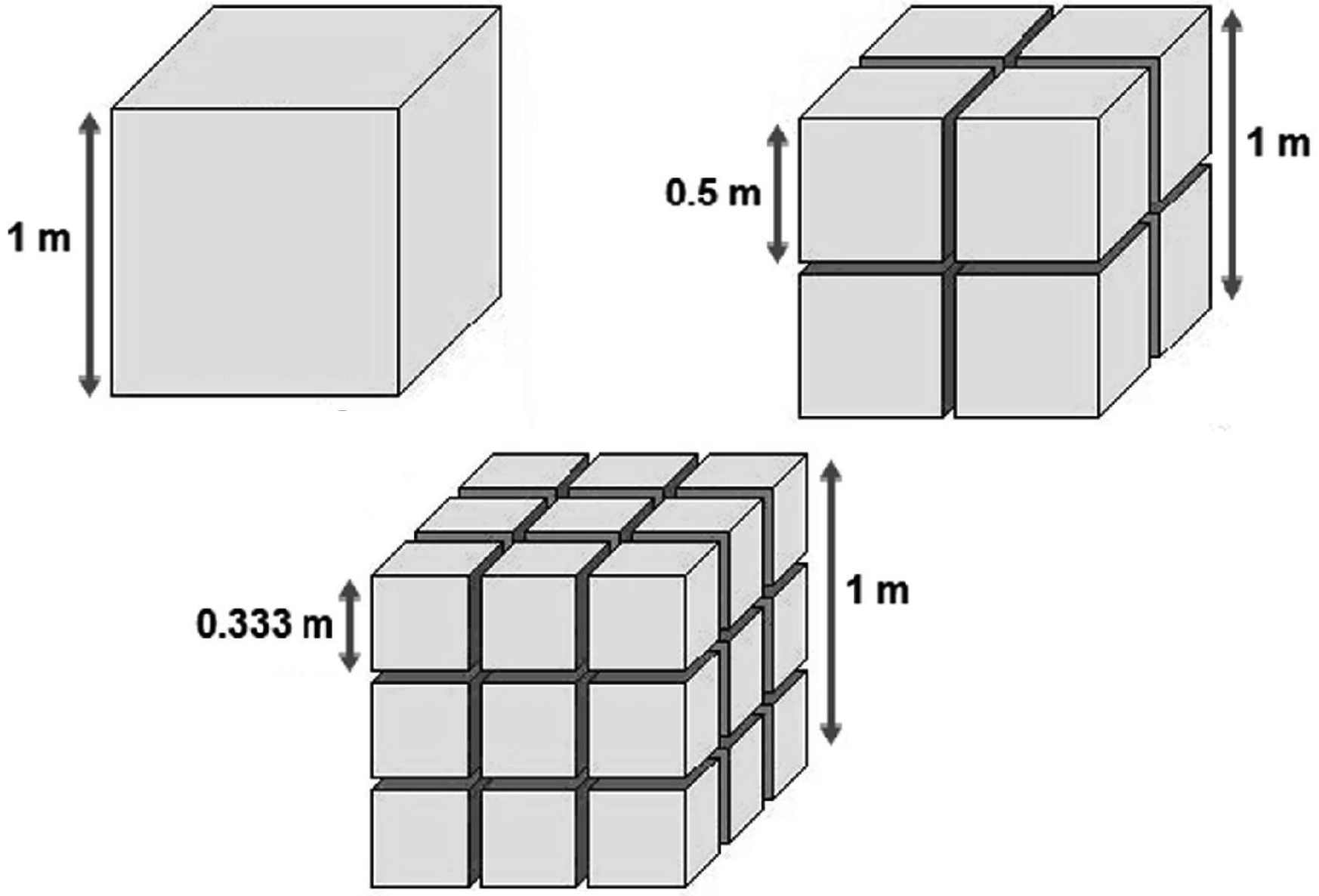


- Inner atoms

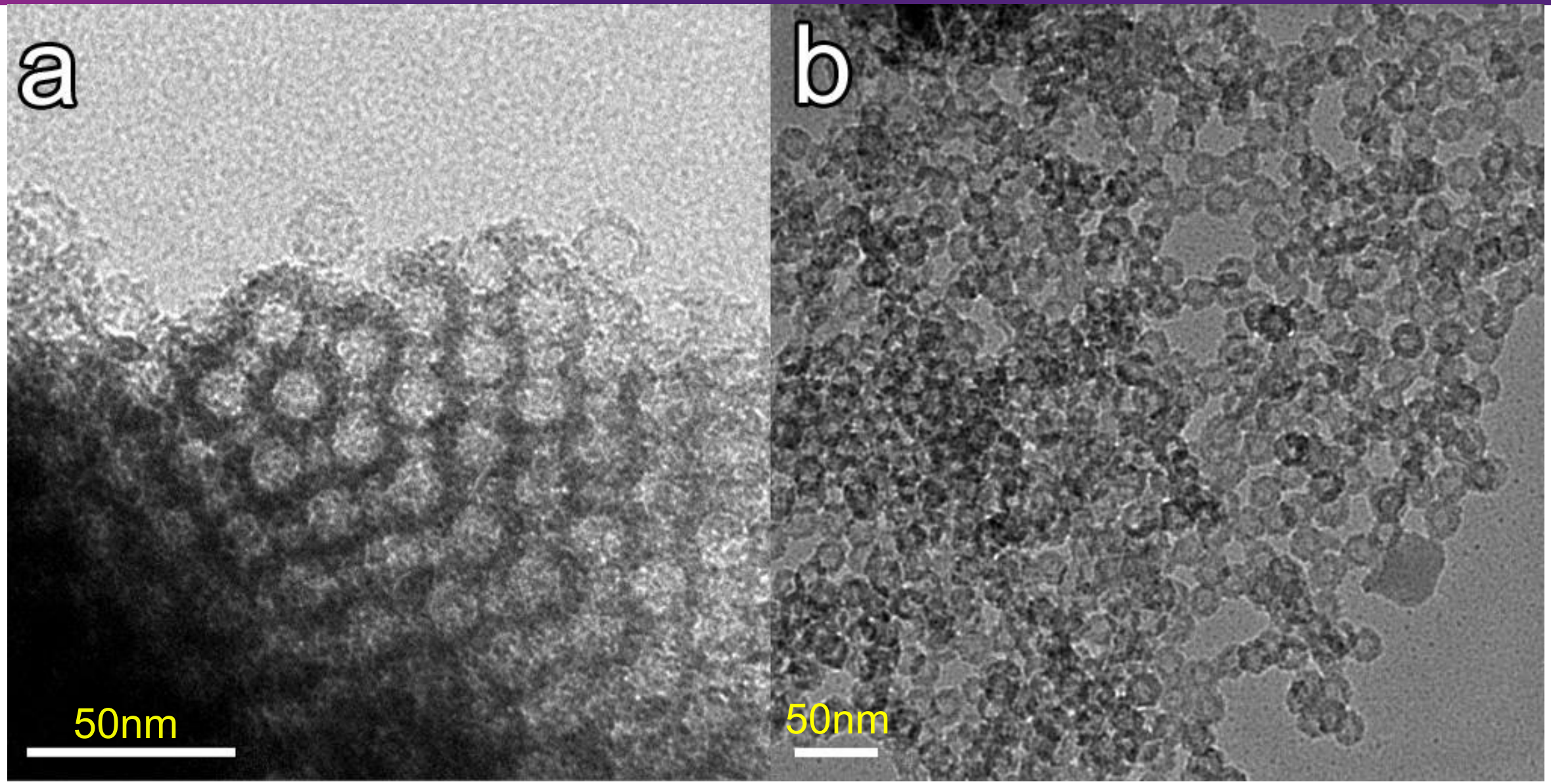
Surface atoms:

- lack of 1 nearest neighbour
- lack of 2 nearest neighbours
- lack of 3 nearest neighbours

Key characteristic of nanomaterials



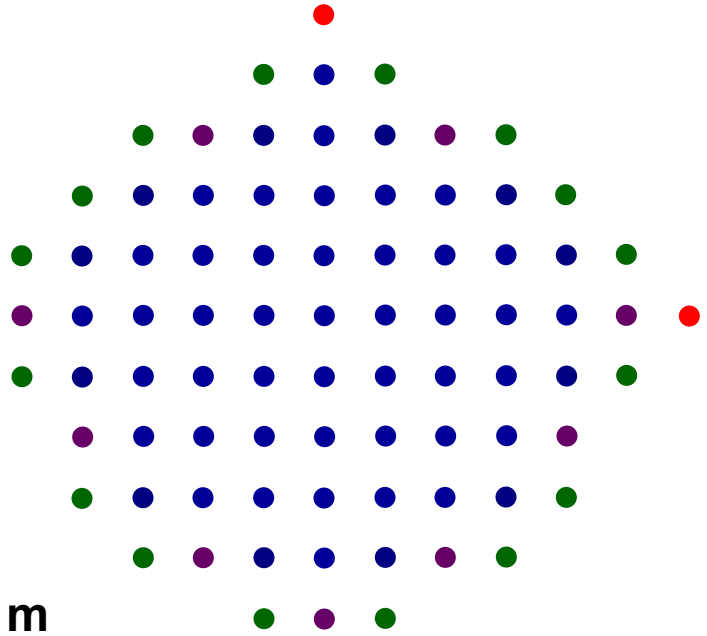
SiO_2 mesoporous structure



The surface area of the mesoporous structure is 400 m^2/g.

Key characteristic of nanomaterials

Significant surface atoms

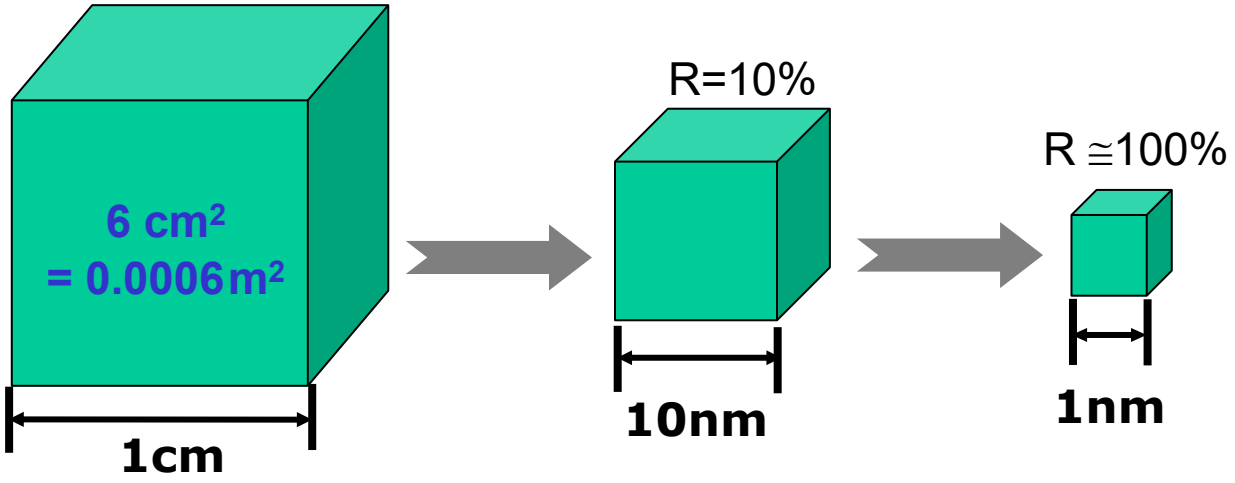


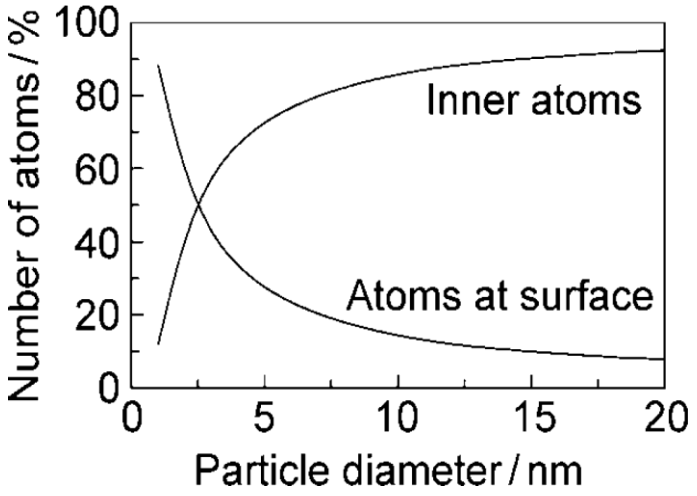
- Inner atoms

Surface atoms:

- lack of 1 nearest neighbour
- lack of 2 nearest neighbours
- lack of 3 nearest neighbours

Nanomaterials: nanometer → 10^{-9} m





The surface area of materials dramatically increases by nanostructure and nanosize.

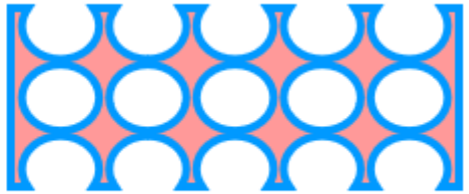
What is a Porous Materials?



Nonporous material

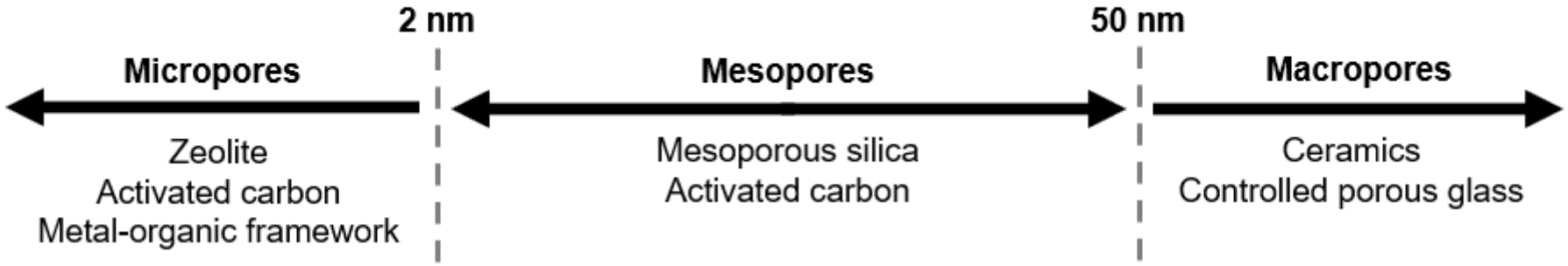
Porous material is a material containing pores

- High specific surface area & Tunable pore size

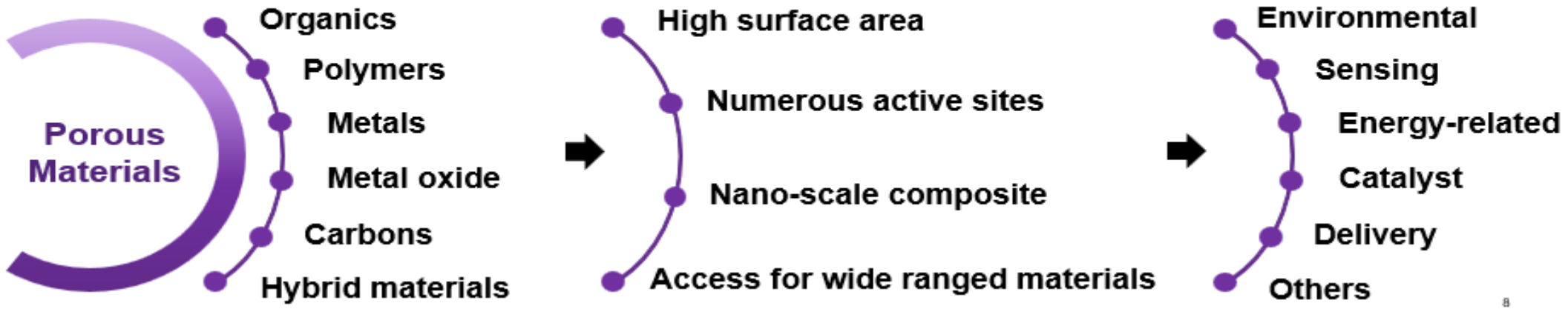


Porous material

Size of Pores (IUPAC standard)



Porous material & applications

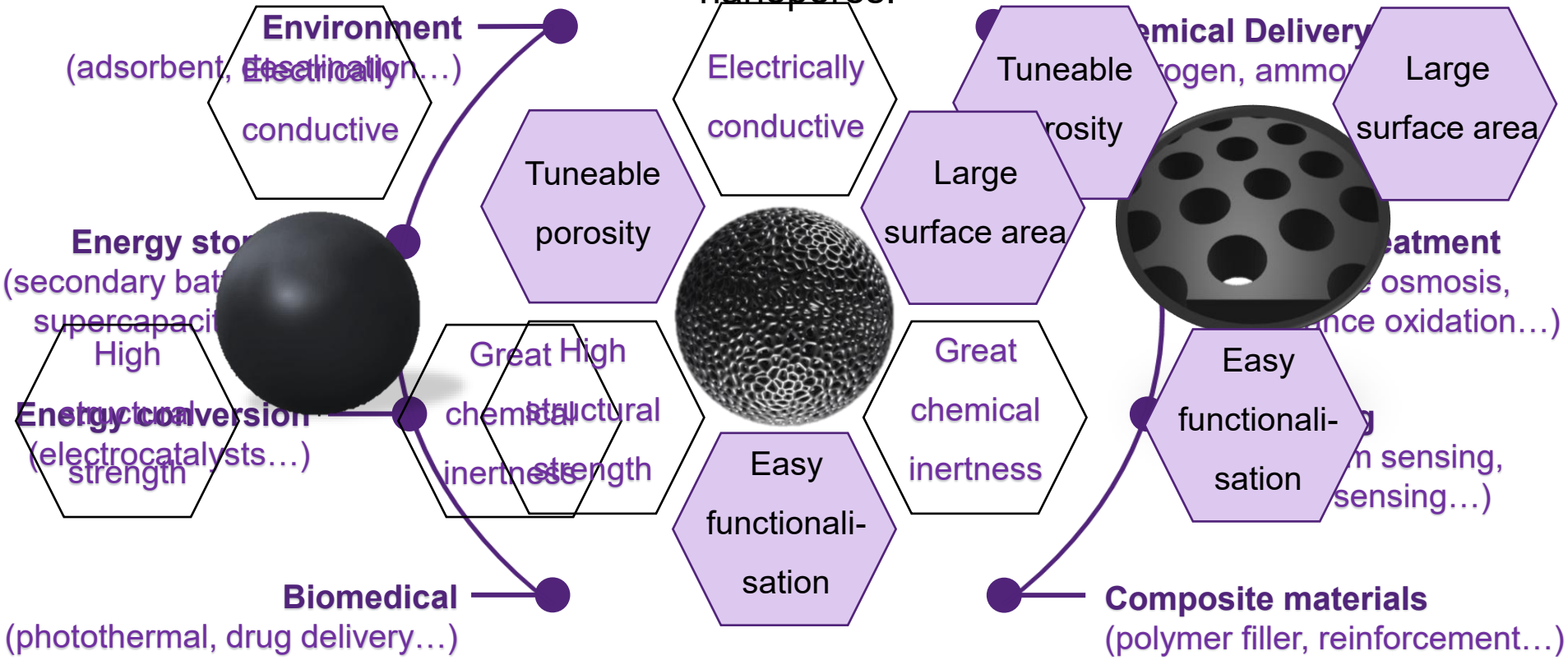


Nano-architectonics

Carbon materials are mainly composed of carbon atoms.

Porous carbon materials are mainly composed of carbon atoms and possess nanopores.

Porous materials are materials that possess pores/voids in nano-scale.



Surface energy

The surface energy (surface tension), γ, is the energy required to create a unit area of *new* surface.

$$\gamma = (\partial G/\partial A)$$

where G is the Gibbon free energy and A is the surface area.

- As particle size goes down, the number of surface atoms increases
- The atoms at corners, edges and surfaces are in the high energy states compared to those in the bulk
- The nanoparticles have high surface energy and high reaction activity

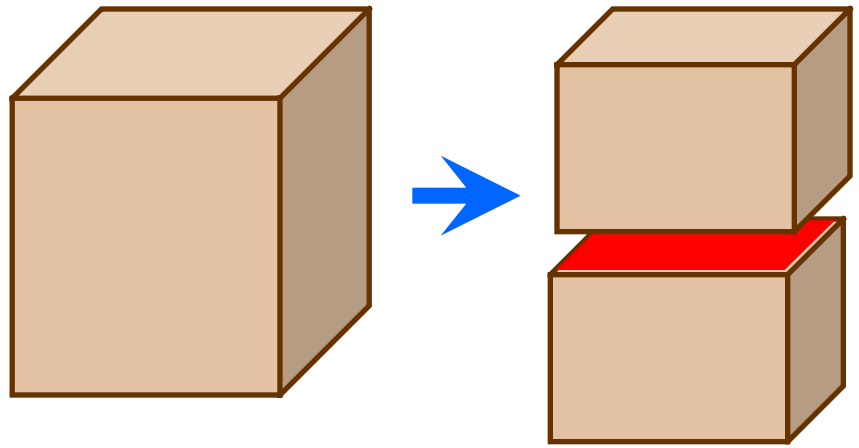
Surface energy

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Considering separation of a rectangular material into two pieces.

Assume N_b is the number of broken bonds, e is the bond strength, and ρ is the number of atoms per unit area on the new surface, the surface energy has the form

$$\gamma = (1/2)N_b e \rho$$



Characteristics of nanoparticles

- large surface area with large amount of atoms in surface
- high surface energy and high surface tension.
- The surface of a nano-sized crystalline particle tends to be faceted
- The faceted planes tends to be the dense plane
- The surface becomes very reactive
  - nanoparticles can be oxidised very easily when exposed in air
  - they tend to be aggregated
- Warning: Metallic nano-particles are very explosive, so that carefully packaging is critical for storage and shipping metallic nanoparticles.

Why Advanced Characterisations?



Synthesising nanomaterials to solve problems that we are facing
e.g., energy, environmental, biosecurity, semiconductors, superconductors…



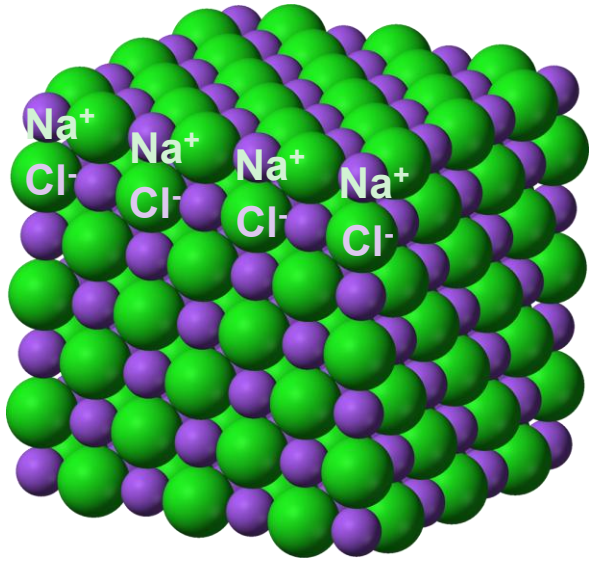
Understanding of the nanomaterials requires advanced characterisations
e.g., atomic compositions, surface topology, crystal structure…



Based on the gained understanding of the nanomaterials by advanced characterisations, we draw some conclusions on materials properties and their roles in certain applications.

Week 2 - What is crystal and crystallography?





Crystal – A crystalline solid material whose constituents (e.g., atoms, molecules or ions) are arranged in a highly ordered microscopic structure, forming a crystal lattice that extends in all directions.

Crystallography – the experimental science of determining the atomic arrangement in crystals.

General tools of crystallography – X-ray diffraction spectroscopy (XRD).

Week 2 - X-ray diffraction (XRD) crystallography

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1. Crystal structures
    1. *Lattices/unit cell/7 crystal systems/14 Bravais lattices*
    2. Symmetries
    3. Crystal directions [ ] and their families < >, as well as their determination
    4. Miller index ( ) of crystal planes and their families { }, as well as their determination
    5. Spacing of crystal planes
    6. Concept of zone axis - Weiss zone law, Interplanar angles, angles between two directions
2. Principle of diffraction
    1. the Bragg law
    2. the structure factor
3. X-ray diffraction
    1. principle of X-ray diffraction
    2. powder X-ray diffraction
    3. applications

- Lab Demo- XRD operation principals and sample characterisation

Week 2 - X-ray diffraction (XRD) crystallography

What information we can get from an XRD analysis?

- Lattice parameters
- Phase identity
- Phase purity
- Crystallinity
- Crystal structure
- Percent phase composition

What information we can not get from an XRD analysis?

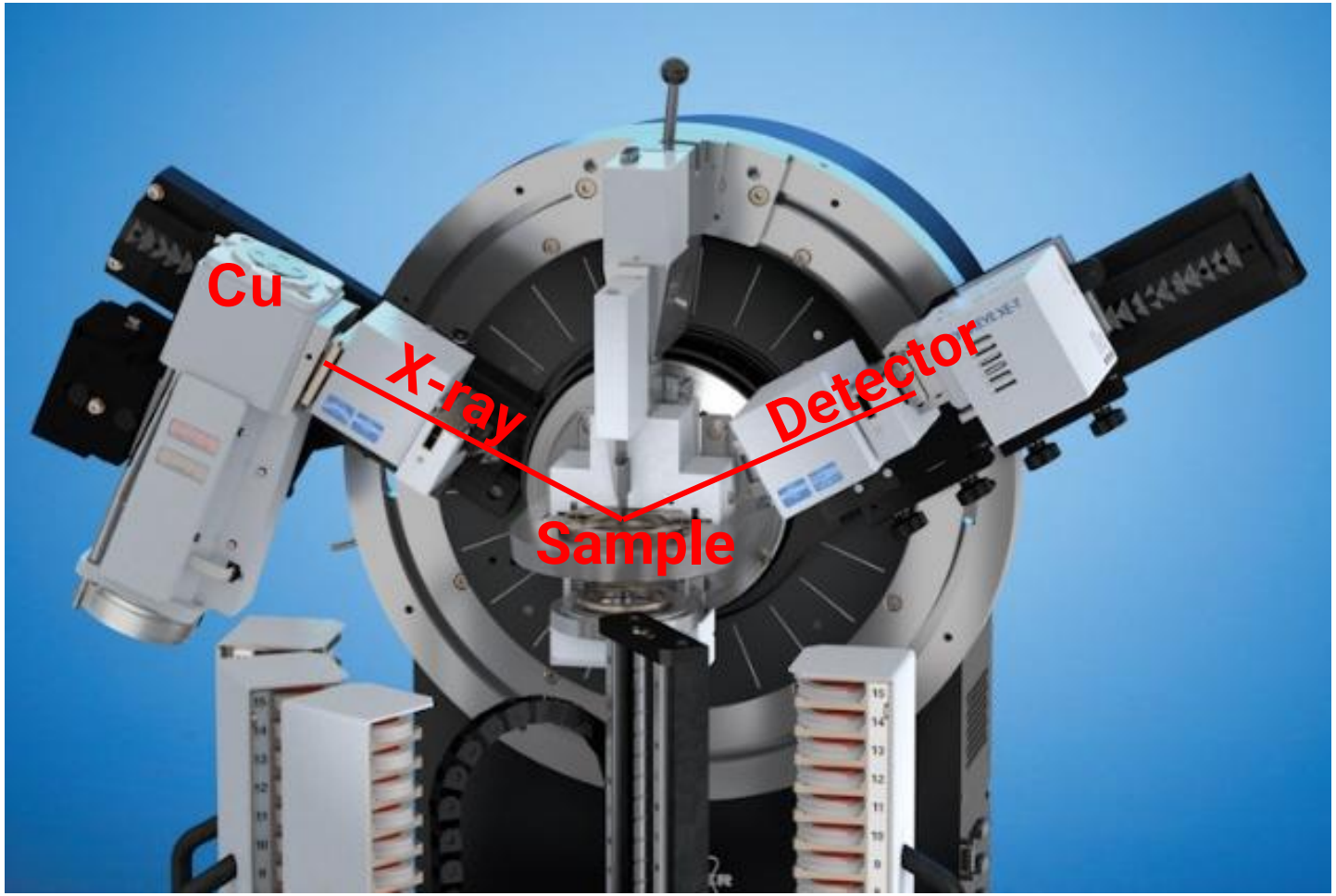
- Elemental analysis -
  - How much lithium is in this sample?
  - Is there iron in this sample
  - What elements are in this sample
- Tell me what this sample is ????
  - Unless you know something about this sample, powder XRD won't have answers !!!

Week 2 - Generation of X-rays



Metal-Ceramic Sealed Tube (Bruker)





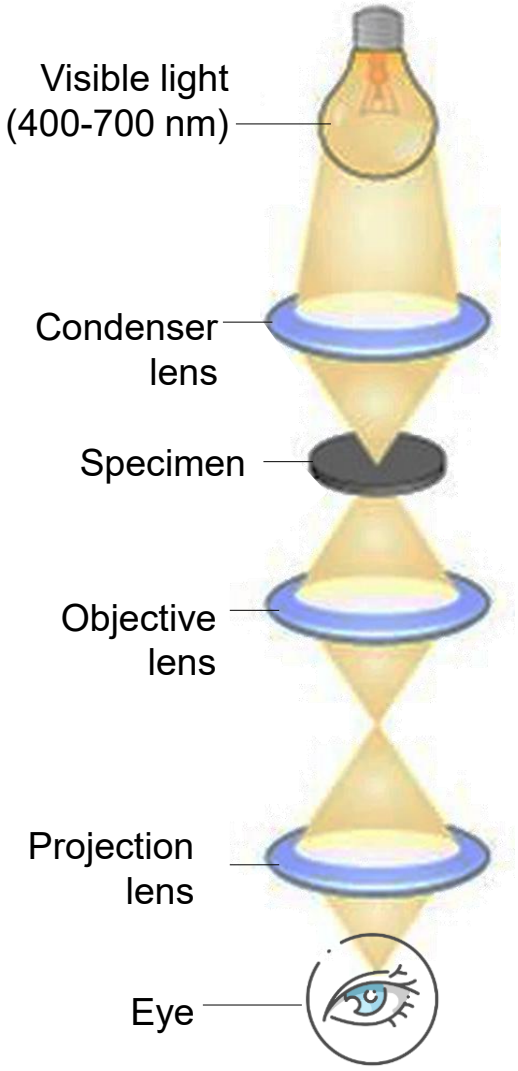
D8 Advance XRD (Bruker)

XRD Simulation: https://myscope.training/XRD_simulator.html

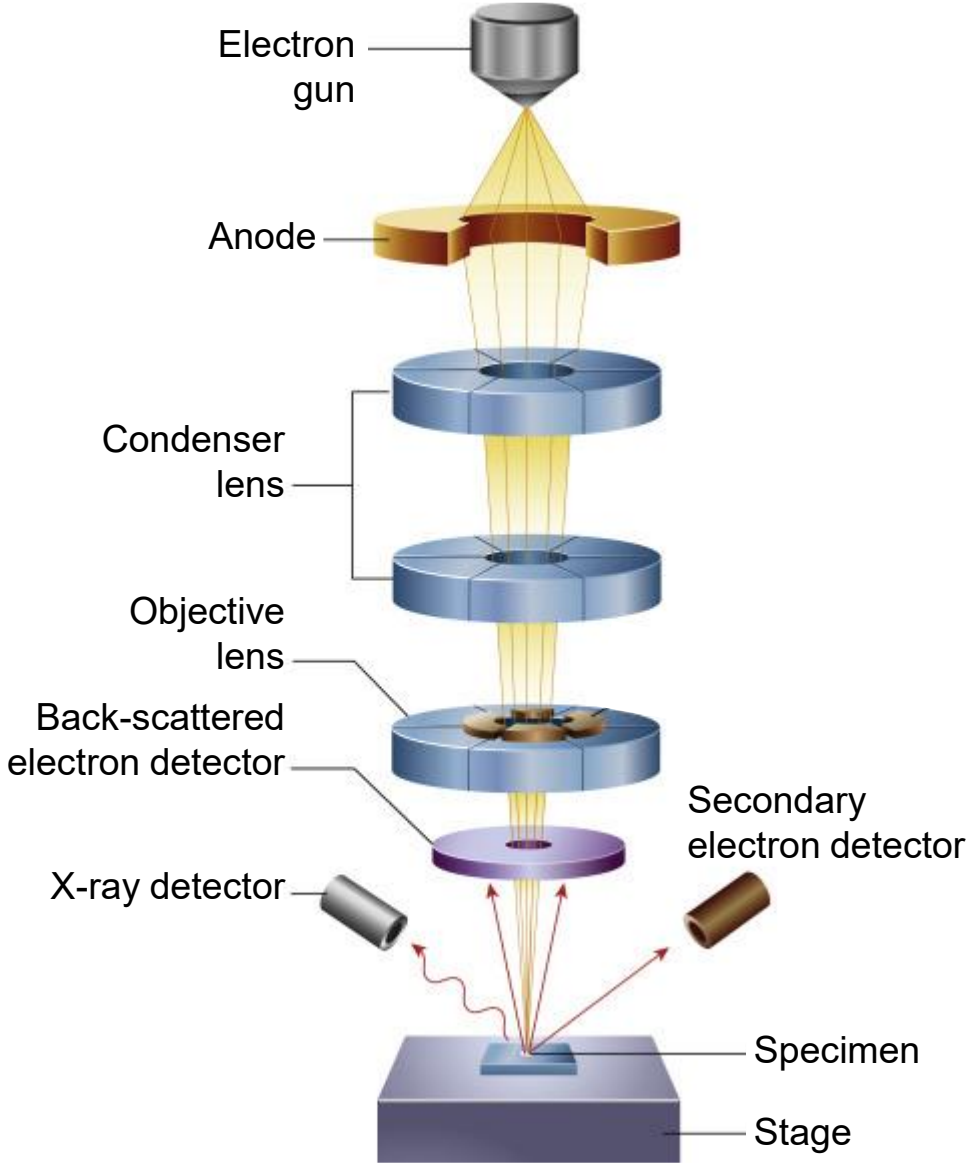
Week 3 - Light Microscope vs Electron Microscope

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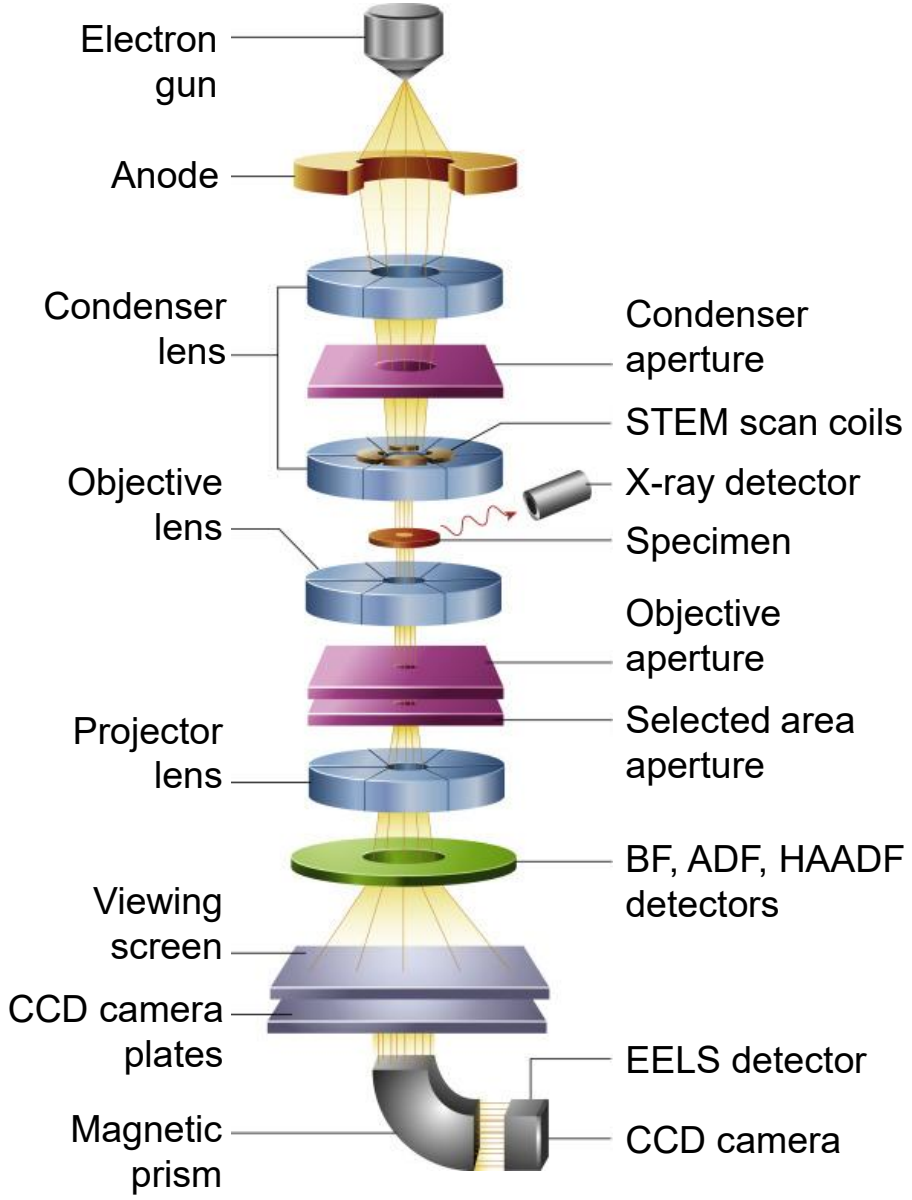
Light microscope



Scanning EM



Transmission EM

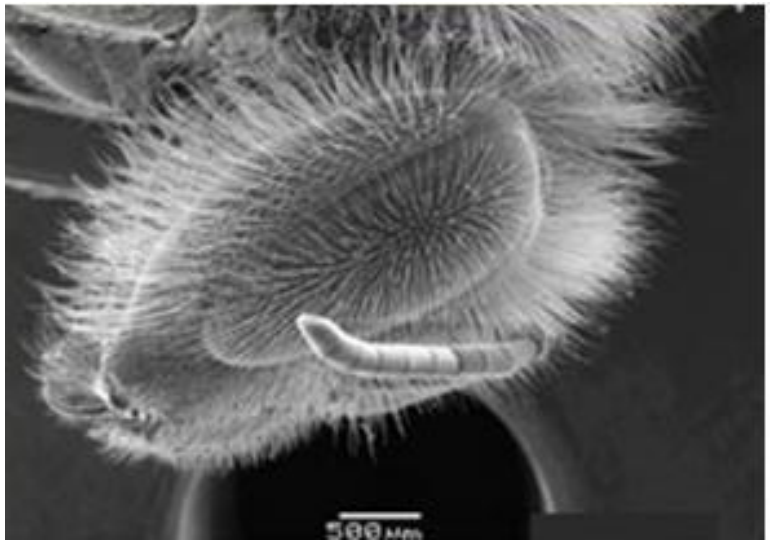


Week 3 - Light Microscope vs. Electron Microscopy

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Light microscope (Colored)



SEM image (Grayscale)

SEM offers…

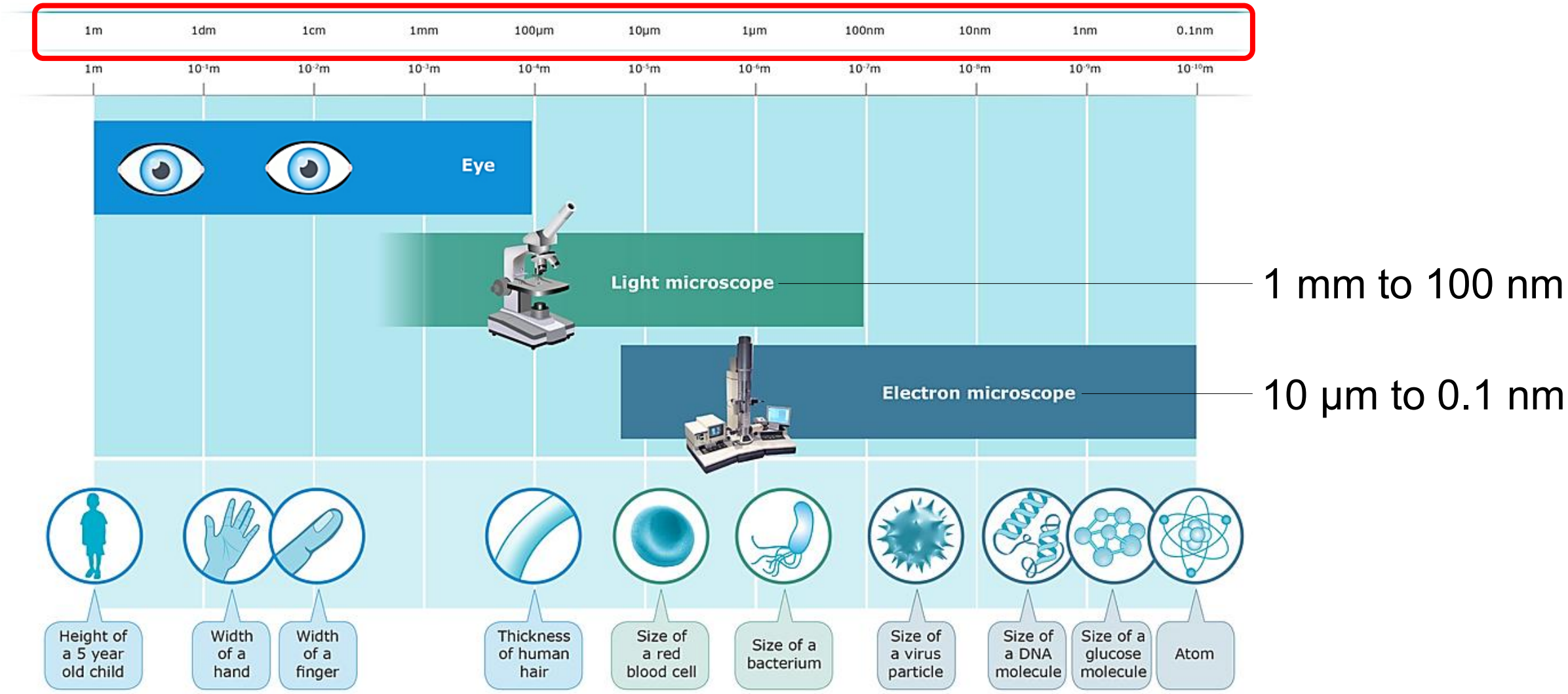
- Much more topographical information.
- Much greater magnification.
- Much higher depth of field.

Why?



Week 3 - Limit of resolution (resolving power)

Limit of resolution (resolving power):
The limit or ability of a microscope to distinguish details of a specimen or sample.



1 mm to 100 nm

10 µm to 0.1 nm

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Week 3 - Electron Microscopy (EM) characterisation

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1. Interactions between electron and specimen - signals
2. Working principles of SEM signals and their applications

SE (Secondary electron)
BSE (Backscattered electrons)
EDS (Electron dispersive X-ray spectroscopy)

1. Operation setting of SEM parameters

Accelerating voltage
Spot size
Aperture
Working distance

1. Applications of SEM
2. Advantages and limitations of SEM

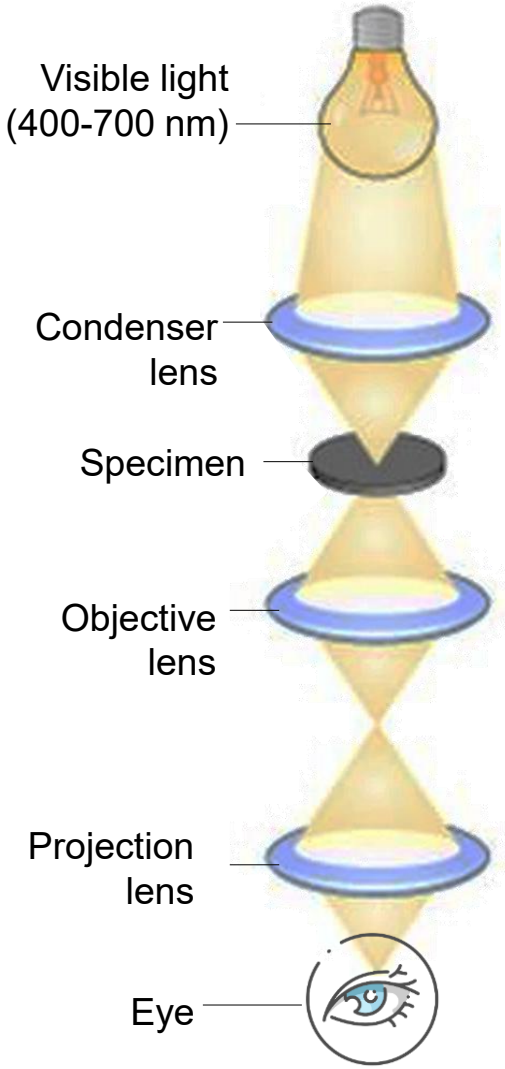
https://myscope.training/SEM_SEM_Basics
https://myscope.training/EDS_Welcome

- Lab Demo- SEM operation principals and sample characterisation

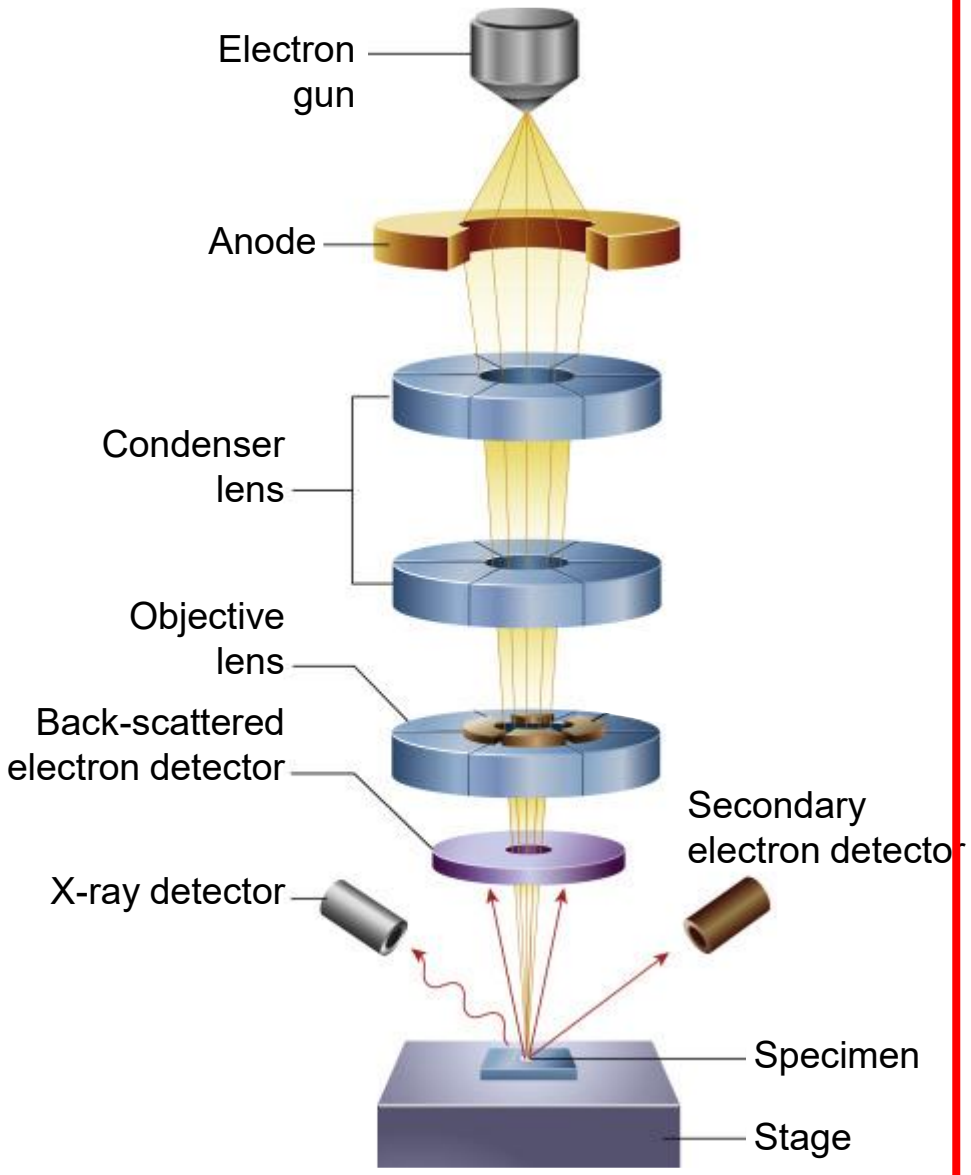
Week 4 - Light Microscope vs Electron Microscope

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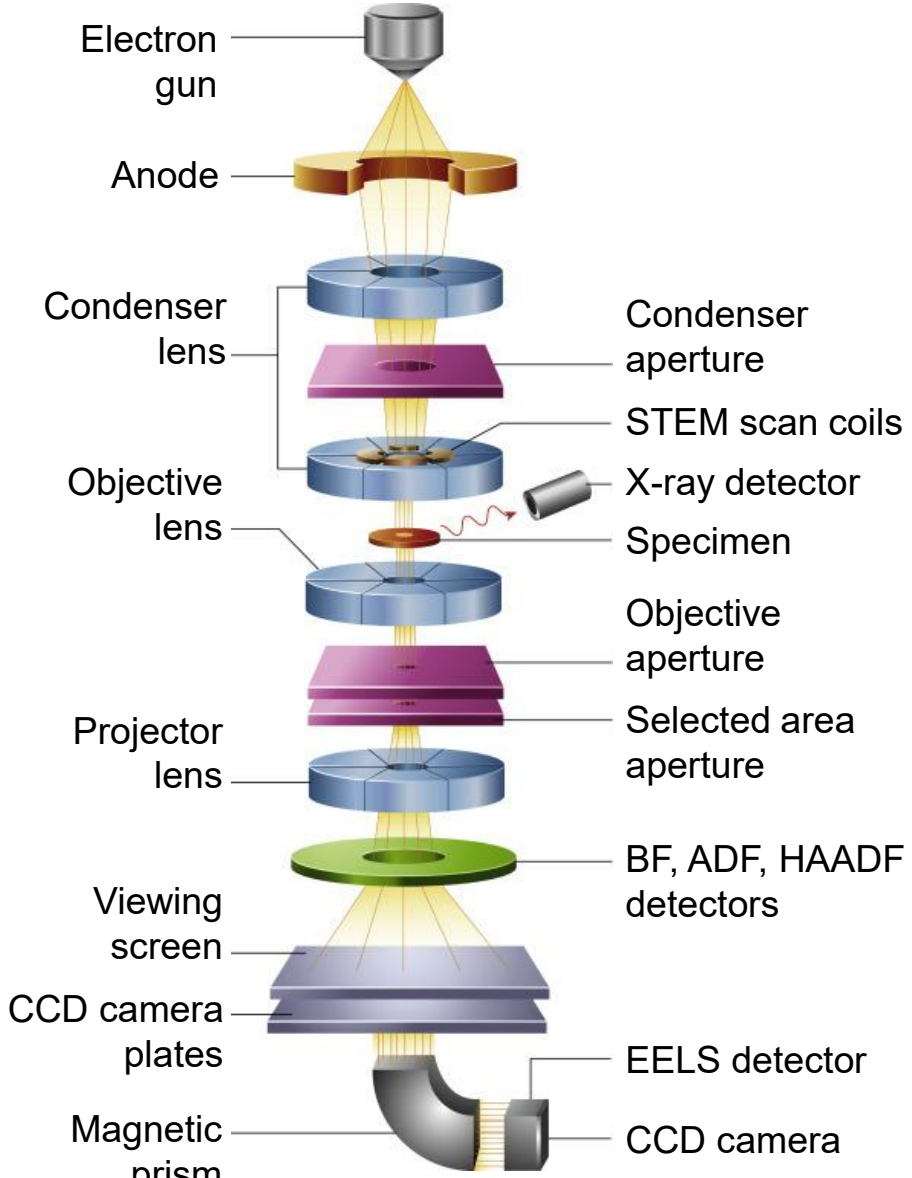
Light microscope



Scanning EM

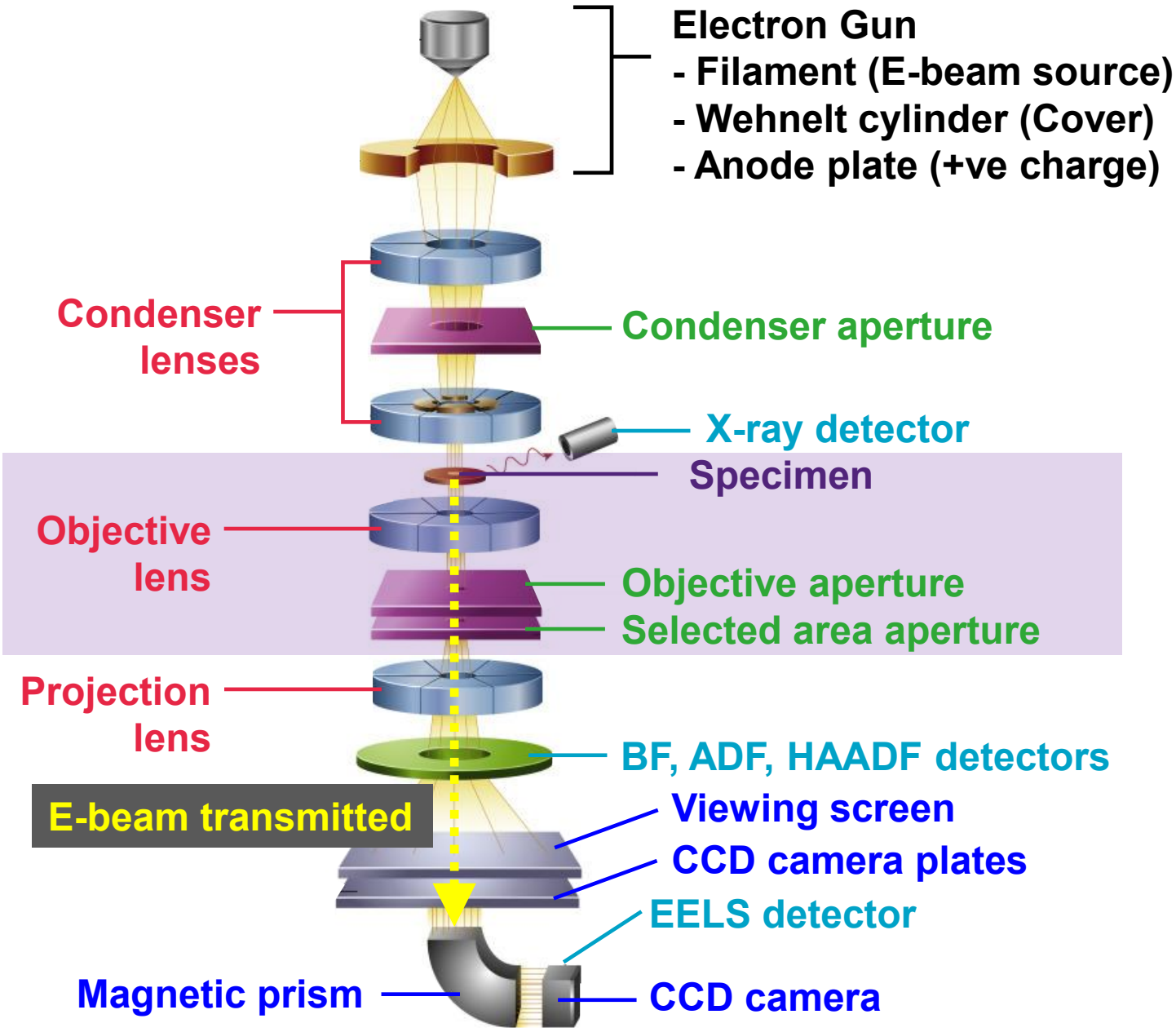


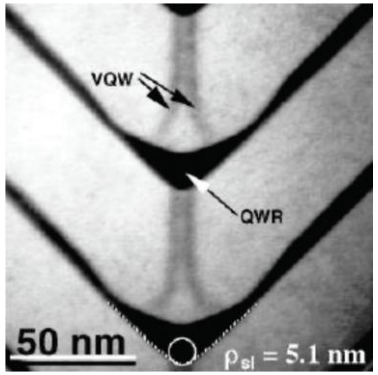
Transmission EM



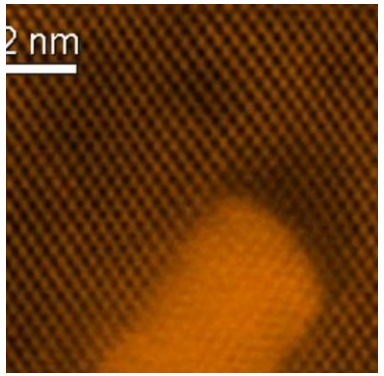
Week 4 -How does TEM work? - The electron gun

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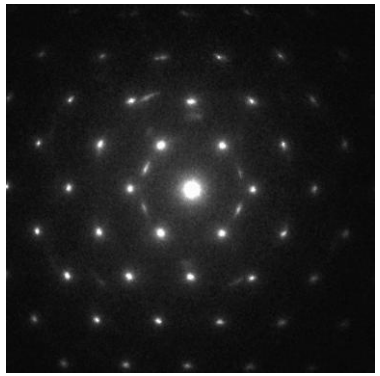




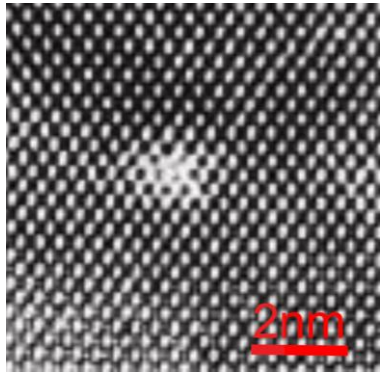
Diffraction contrast



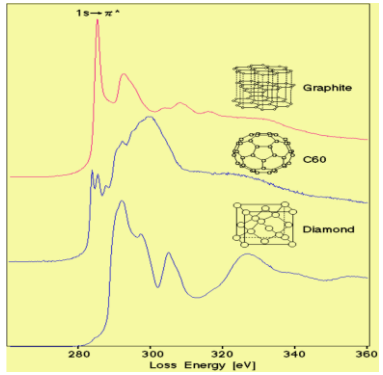
STEM imaging



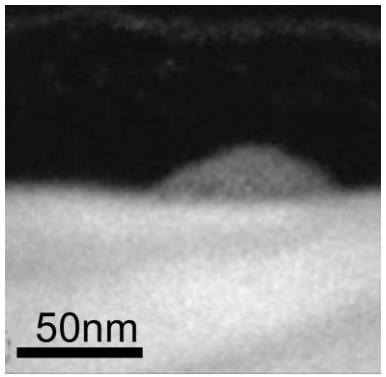
Electron diffraction



HRTEM imaging



EELS spectroscopy



Energy filtered TEM

Week 4 - Electron Microscopy (EM) characterisation

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1. Structure and Principle of a TEM
2. Electron diffraction
3. Diffraction contrast (Imaging formed by single diffraction spot)
4. Phase contract (HRTEM) (Imaging formed by multiple diffraction spots)

   TEM specimens must be thin and well orientated

1. Analytical TEM

Characteristic X-rays

Electron energy lose spectroscopy (not only chemistry, but also bonding)

STEM (Z contrast)

1. Applications of analytic S/TEM

- Lab Demo- TEM operation principals and sample characterisation

Lecture contents and Related Assessments

X-ray diffraction (XRD) - Week 1, 2

- Lecture 2
- 2Lab demonstration

Scanning electron microscope (SEM) **) - Week 3, 4**

- Lecture 3
- 2Lab demonstration

Transmission electron microscope (TEM) **) - Week 5, 6**

- Lecture 4
- 2Lab demonstration

Final examination

To be consolidated to Lab Engagement + Lab report (20%)

Week 5 - Nanomagnetism and mag nanocomposite

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- What are magnets and how they work
- Origin and fundamentals of magnetism
- Magnetic materials and their types
- Temperature effect and magnetic domain
- Magnetic hysteresis
- Magnetism and super-paramagnetism
- Synthesis of Magnetic Nanoparticles.

  - Lab demo on the synthesis ofn magnetic nanostructures

Week 6 - Characterisation/analysis and application of mag nanocompo

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Parameters influence magnetic properties of nanoparticles/nanocomposites

Effect of annealing temperatures on magnetic phase transition and magnetic properties of nanoparticles

The particle size effect on magnetic properties and determination of critical size.

The particle shape effect on magnetic properties.

The density/porosity effect on magnetic properties of nanoparticles.

Nano Machineries – Magnetic drug delivery applications

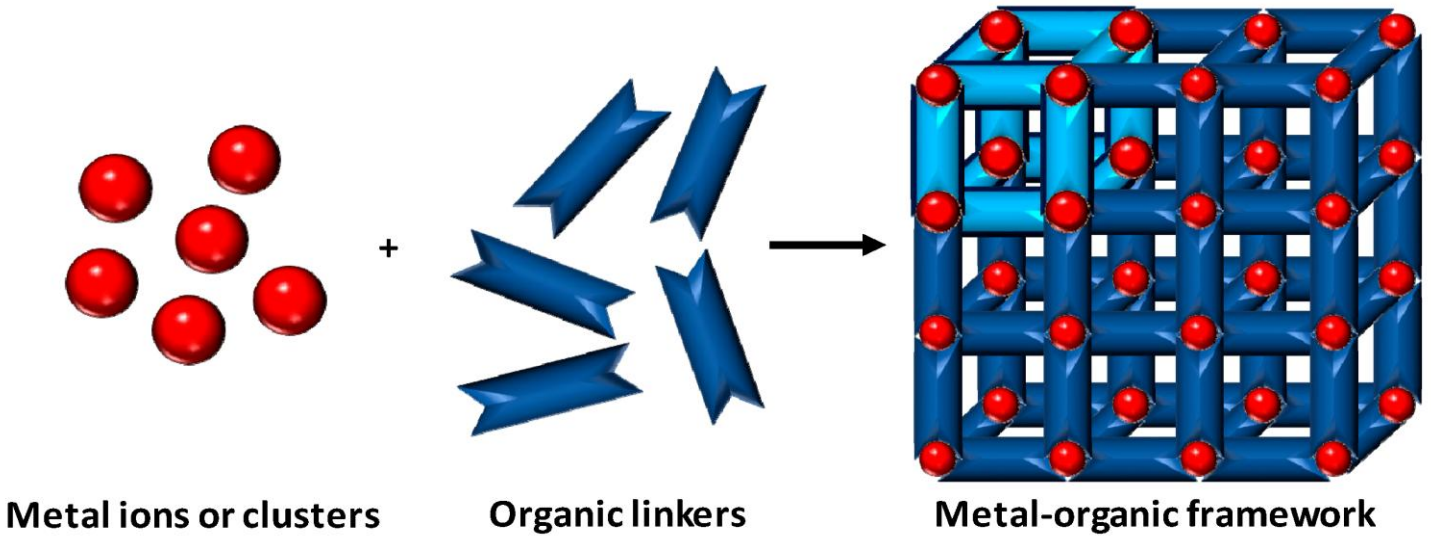
- Laboratory – Electrochemical characterisation of magnetic nanocomposites/analysis

Week 7 - Porous Nanocatalysts Waste water treatmen

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METAL-ORGANIC FRAMEWORKS (MOFs)

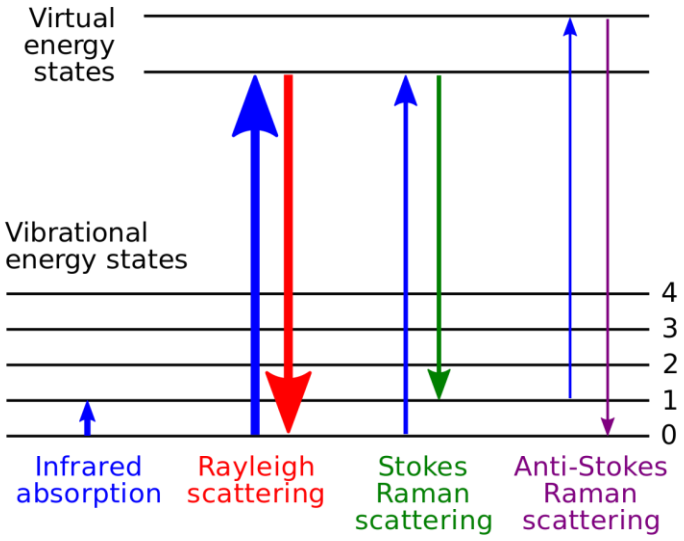
Crystalline, porous materials formed by the simple coordination of metal centres and organic linkers/ligands

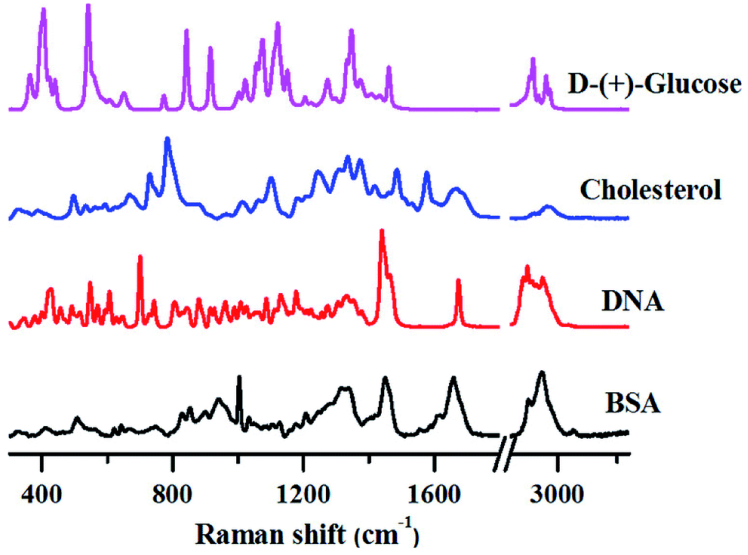


- Demo- Nanocatalysts (MOF) for waste water treatment

Week 8 -Optical properties of nanocomposite and their applications

Principles of Raman Spectroscopy: Raman spectroscopy is a type of spectroscopic techniques that determines the vibrational modes of the molecules. It relies on inelastic scattering of photons.





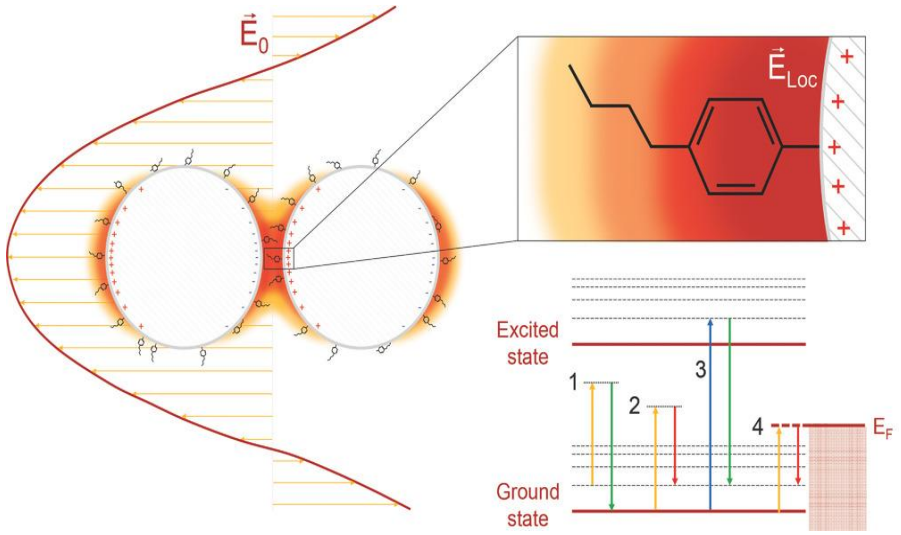
Vibrational and rotational frequencies of a molecule are characteristics of a molecule.

Kuhar *et. al. RSC Adv.* **8**, 25888-25908 (2018)

- Laboratory – metallic nanocomposites with optical properties for SERS based biosensing for pesticide detection

Week 9 –Bi-tri metallic nanocomposite film and their optical applications

Signal Enhancement mechanism of Raman Spectroscopy using nanomaterials



Electromagnetic enhancement:

- Incident light on the nanomaterials creates localised surface plasmon resonance
- Plasmon oscillates
- Scattering of the light increases with perpendicular oscillations

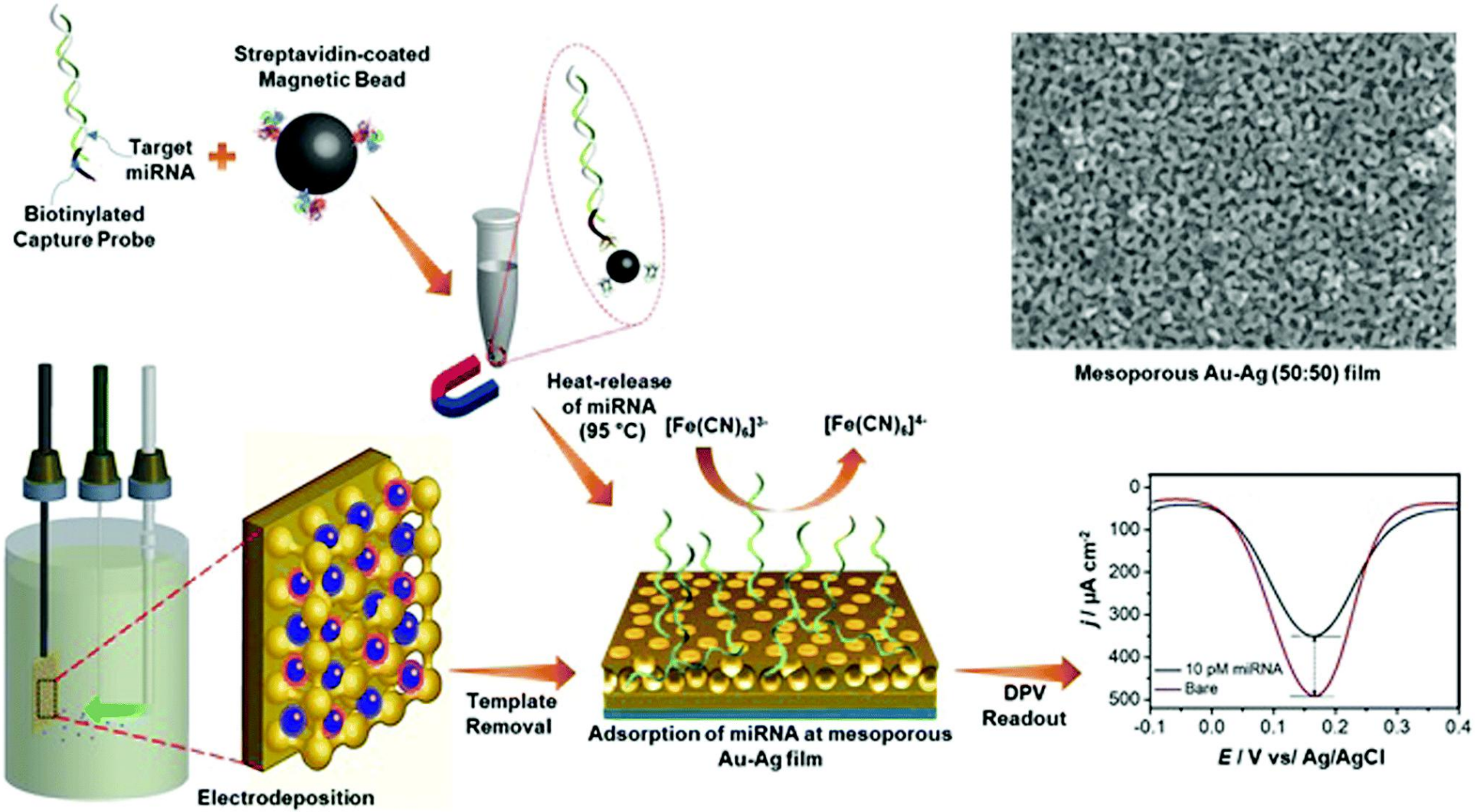
Chemical Enhancement:

- Incident light on the nanomaterials creates chemisorption of the molecule on the nanomaterial surface
- Charge transfer occurs
- Scattering of the light increases due to the charge transfer phenomena

Balcytis *et. al. Adv.Optical Mater.* 6, 1800292 (2018)

- Laboratory – metallic nanocomposites with optical properties for SERS based biosensing for pesticide detection

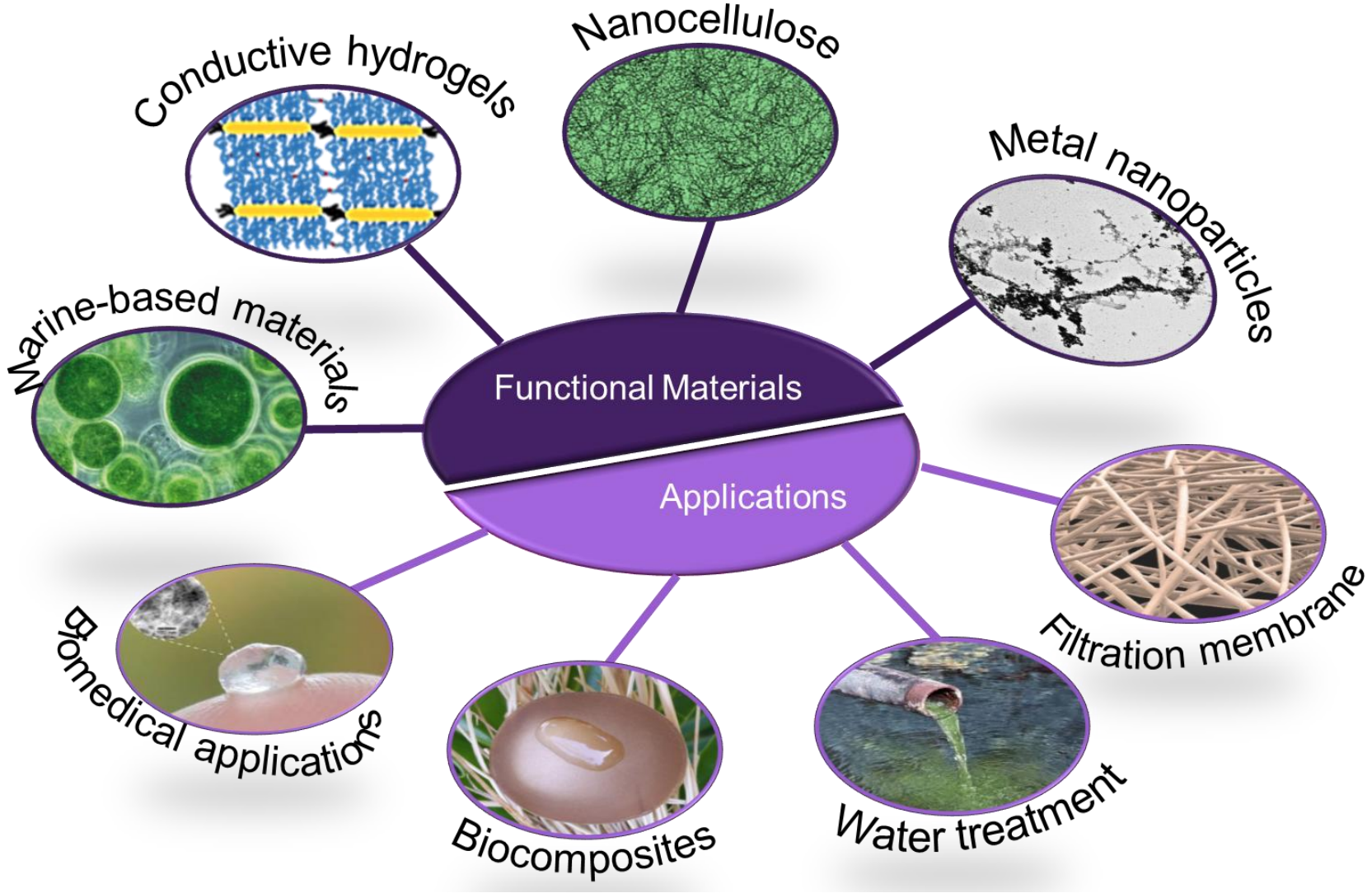
Week 9 –Bi-tri metallic nanocomposite film and their optical applicatio



https://pubs.rsc.org/en/content/articlelanding/2020/tb/d0tb02003f

- Laboratory – metallic nanocomposites with optical properties for SERS based biosensing for pesticide detection

Week 10 - Australian biomass derived nanocellulose



Online class and lab prac due to the public holiday

Week 11 – Presentation preparation and assessment

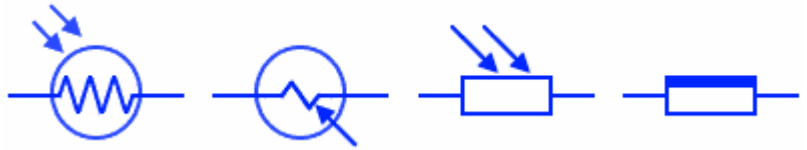


Week 12 – Semiconductor based photodetector application

What are photodetectors?

A photodetector is a variable resistor whose resistance depends on the light intensity of incident light. So, this is also called as *Light Dependent Resistor* (LDR)/*photoresistor*. It works on the principle of *photoconductivity*.

Symbol:



Example:

CdSe and CdS, used in *light sensing circuits* such as automatic street lights, solar lights, fire alarms, burglar alarms, etc.

Advantages:

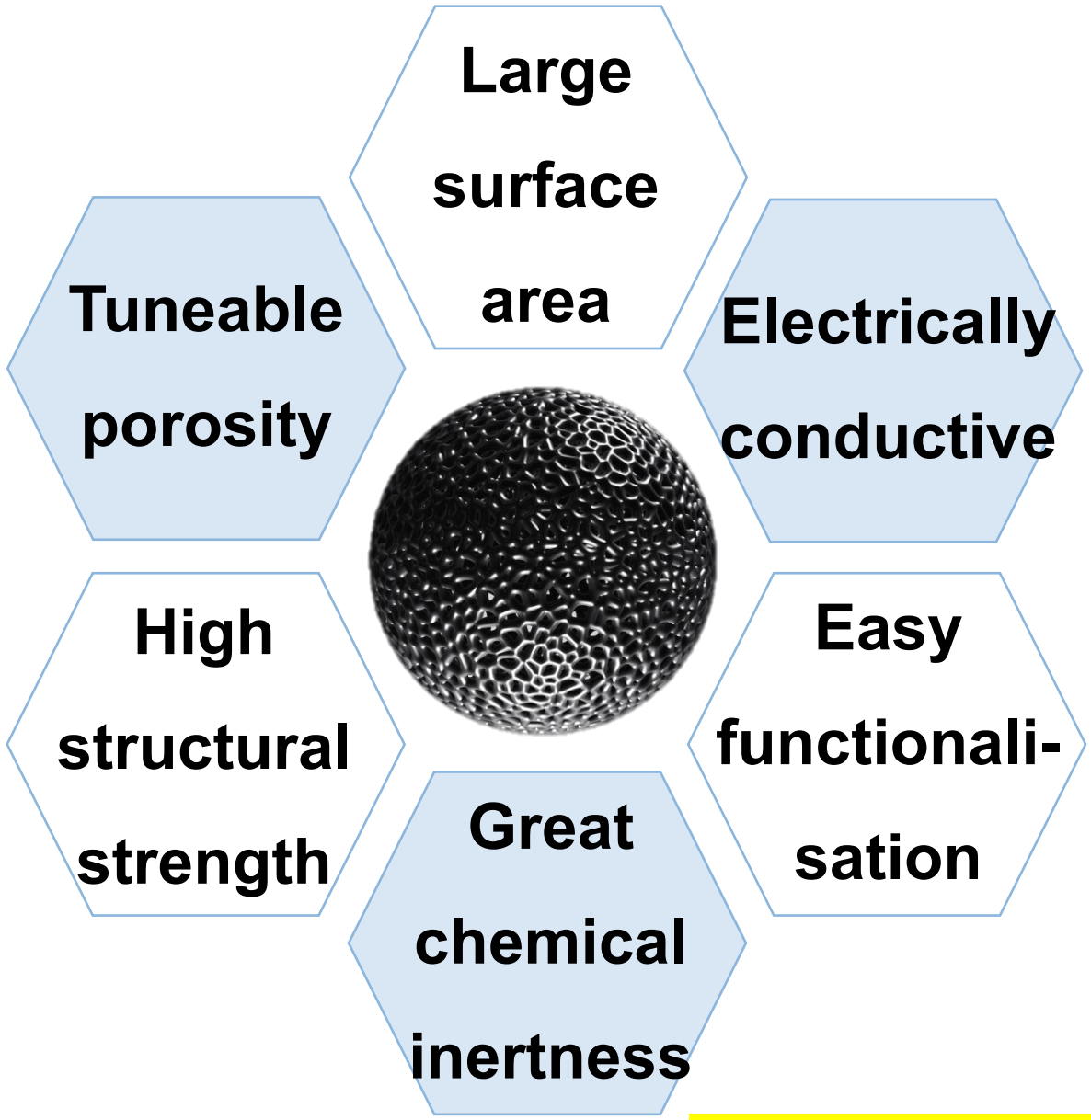
- Non-linear response to the intensity of light.
- It operates in either biasing. (It can conduct in both directions.)
- It is made of a single N-type semiconductor.
- Cost-effective and durable.
- Has a simple design and is easy to manufacture.

Disadvantages:

- It has a slower response time.

- Laboratory – photodetector demonstration

Week 13 - NANOPOROUS CARBONS



- Highly suitable for energy–related applications.
- Examples are secondary battery, supercapacitor, water-splitting electrocatalysis, hydrogen fuel cell, etc.
- It can be obtained from various materials such as biomass, porous polymers, by-products of oil refinery, etc.

▪Online Lab prac demo porous carbon synthesis and application in supercapacitors.

Week 13 – Final Exam Discussion

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Q/A on final exam and discussion

MATE 6301 (Laboratory)

Deposition and Characterization of Mesoporous Gold Thin Films

Electrochemical Deposition

Electrochemical deposition is a process in which metallic ions can solidify and deposit on the cathode surface if an adequate amount of electric current flows through the electrolyte solution. The charged ions in the electrolyte are produced by dissolving metallic salt in water. Using the electrodes, an electric current is introduced into the electrolyte during the electrochemical deposition procedure.

Operating Principle

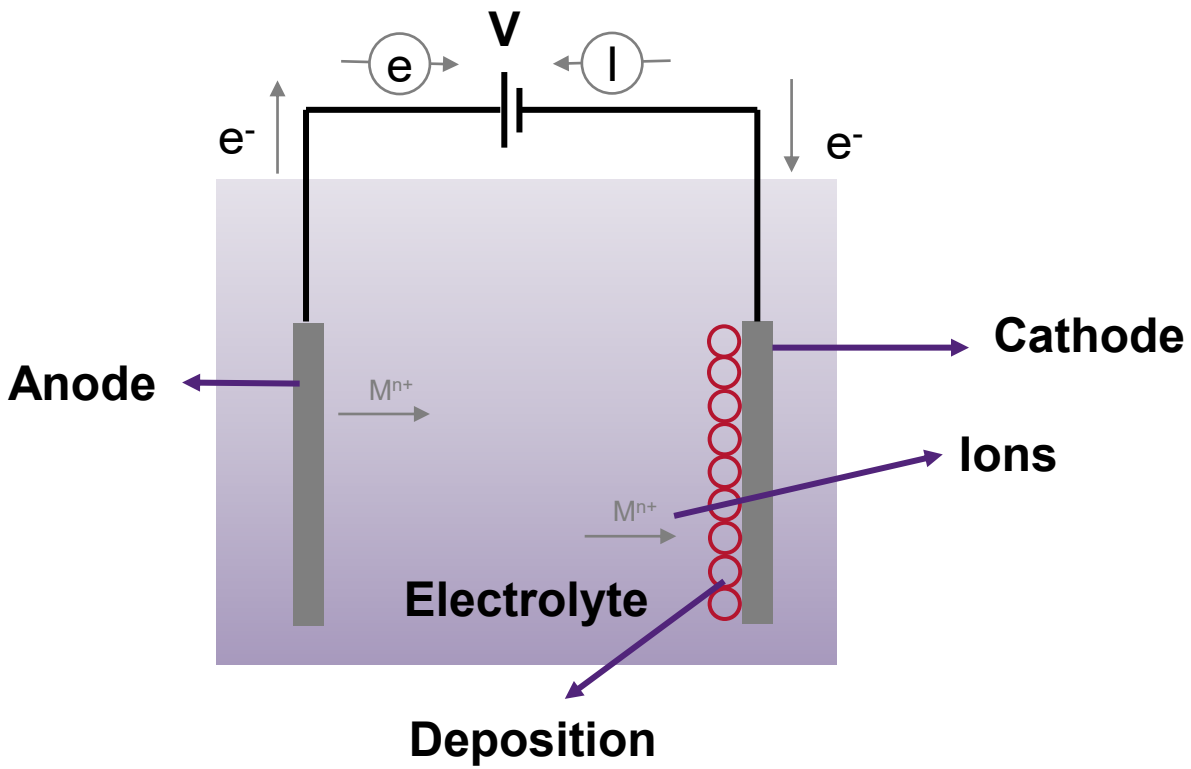
- Reduction of metal ions in an electrolyte solution onto a conductive surface through the application of an electric potential.

- The general equation for the electrochemical deposition of a metal (M) from an electrolyte can be represented as follows:

$$M^{n+} + ne^- \rightarrow M$$

where,

n: oxidation state, e: electron.



The electrochemical deposition process follows:

1. **Migration** – the hydrated metal ions in the solution migrate towards the cathode under the influence of applied current as well as by diffusion and convection.
2. **Electron transfer** – at the cathode surface, a hydrated metal ion enters the diffused double layer where the water molecules of the hydrated ion are aligned. Then the metal ions enter the Helmholtz double layer where it is deprived of their hydrate envelope.
3. The dehydrated ion is **neutralized and adsorbed** on the cathode surface.
4. The absorbed atom then migrates or diffuses to the **growth** point on the cathode surface.

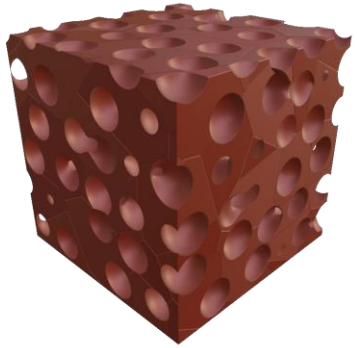
[1]

The difference:

Conventional thin films

- Low surface roughness
- Low surface area
- Less surface defects
- No change in the refractive index
- No tunable functionality
- More material is used
- Low surface sensitivity
- Micro to nano crystalline grains



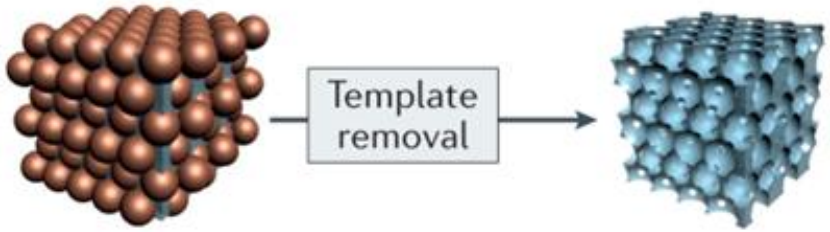


Porous thin films

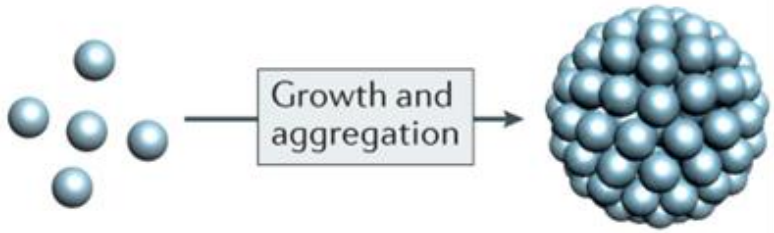
- High surface roughness
- High surface area
- More surface defects
- Tunable refractive index
- Tunable functionality
- Less material is used
- High surface sensitivity
- Nano crystalline grains

How are these porous nanostructures formed?

Template-assisted



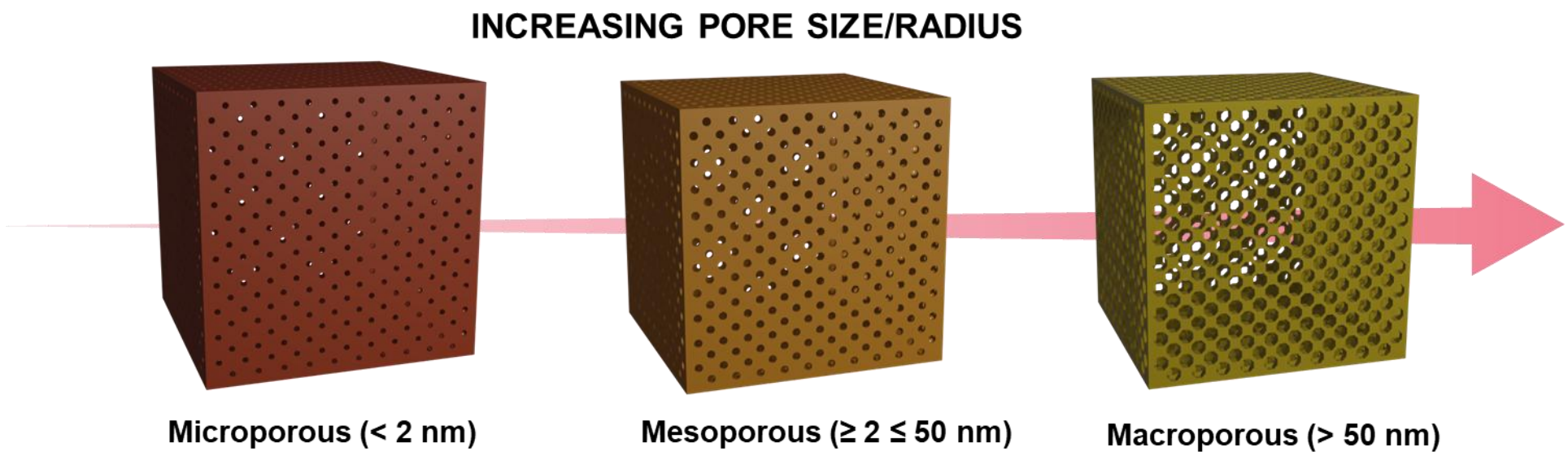
Template-free



Porous thin films?

Introducing defined void spaces also referred to as “pores” at the bulk of the film distributed symmetrically or asymmetry in a material system.

Classifications of porous materials:



Importance of mesoporous thin film?

High surface area, versatile surface functionalization, and bicontinuous percolation paths through a broad library of pore arrangements which results in improved sensitivity and detectivity. Surfaces in porous materials are more diffusive than specular. This can minimise the net reflective loss in functional materials.

Check the announcement and group formation for the lab demo starting from week 1 Friday 8-12pm

Thank you and all the best!